Stock Focus

The Lottery Corporation (TLC-ASX)

E&P

Thursday, 3 August 2023

TLC - burger with The Lott

Recommendation: Neutral

Last price	**AUD5.23**
Valuation	**AUD5.79**

TSR	
Upside to valuation	10.8%
Dividend yield	3.1%
Expected total SH return	13.8%

TLC-ASX share price history

Source: E&P Research, IRESS



Trading Data	
Last price	AUD5.23
Valuation	AUD5.79
12 month range	AUD3.92 - AUD5.32
Market cap ($m)	$11,641
Shares outstanding (m)	2,225.8
Free float (%)	100.0
12 month return (historical)	14.4%

Earnings revisions

	FY23	% chg	FY24	% chg
Revenue	3,531.0	245.3%	3,843.5	215.8%
EBITDA	727.3	160.0%	837.6	141.8%
Adj. EPS	14.3	(76.1)%	19.4	(75.0)%
DPS	0.2	(59.5)%	0.2	(70.6)%
Val	**AUD5.79**	**29.92%**		

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Restarting the game

We re-launch coverage on The Lottery Corporation (TLC) with a Neutral recommendation and $5.79/sh valuation. Coverage transfers to Paul Mason and Sam Bradshaw as part of this report. We are more positively inclined than not on TLC, with a valuation and dividend yield that implies low-teens upside, but in the short term we believe consensus may lower revenue and EBITDA expectations.

As part of this report we have upgraded our entries analysis and monte carlo simulation capabilities for TLC's jackpot games. This report runs through the following major elements:

- Long term lotteries entry growth
- Estimating lottery entries
- Modelling Powerball and developing a Monte Carlo simulation framework for a 50th percentile view on revenues

The report is reasonably heavy on maths and statistics, and ends up with a 50th percentile valuation on TLC. In practice this analysis may be difficult for investors to replicate given the data gathering and processing required, but hopefully presents as robust. A quick shout out to Sam Bradshaw, the associate involved in building this report, whose efforts in building out the technical aspects of this report were above and beyond.

Changes to forecasts and valuation

We have increased our EBITDA and NPAT forecasts for FY23-25 by +7.0%/ +12.6%/+9.2% and -11.8%/+5.8%/+3.4% respectively. Our valuation has increased to $5.79/sh on transfer of coverage to Paul Mason and Sam Bradshaw.

Our view

We see TLC as a high quality business with regulatory protections and a relatively resilient underlying revenue base (probability adjusted). At the right settings of consensus estimates and valuation we see TLC as a good long term investment with predictable and attractive returns potentially on offer. For now we are a little worried about a short term consensus re-base, but otherwise, while running with a Neutral recommendation, are generally positively disclosed towards the company.

Earnings Forecasts

Yr to Jun (AUD)	22A	23E	24E	25E
EBITDA	606.1	727.3	837.6	866.0
Adjusted Net Profit	342.7	319.3	432.8	459.0
Adj. EPS (c)	15.4	14.3	19.4	20.6
PER (x)	**34.0**	**36.5**	**26.9**	**25.4**
Yield (%)	**0.0**	**3.3**	**3.1**	**3.1**
EV/EBITDA (x)	**22.6**	**18.7**	**15.9**	**15.2**
Valuation (blended)				**AUD5.79**

Source: E&P Research

Stock Focus

The Lottery Corporation

Investment Thesis and report overview

The Lottery Corporation (TLC) is a generally solid portfolio company for any investor willing to invest in the gaming sector. The company should generate consistent returns through the 'probability cycle' at its 50th percentile jackpot outcomes, and is positively exposed to population growth as well as having the ability to grow from changing games over time (which is very hard to model and value).

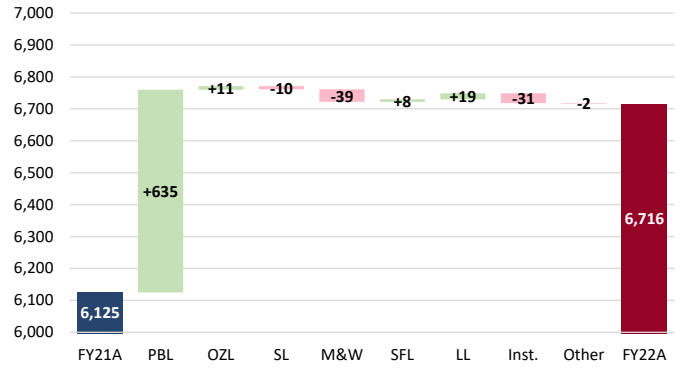
We initiate with a Neutral recommendation and $5.79/sh valuation. We believe the market is slightly over-estimating 50th percentile revenues for the company presently and may re-base slightly downwards during reporting season, albeit we also believe the market may be underestimating the benefits of the recent changes to retail commissions. So we are expecting some figures to get moved around this reporting season.

In terms of our view on the company itself as opposed to market expectations, we see TLC as a solid investment, with general regulatory monopolistic or oligopolistic characteristics, resilience to economic cycles, and a history of business development driving growth. Given the 13.7% expected shareholder return in our valuation, this is presently on the cusp between Neutral and Positive for us, and any significant draw-downs in share price would be viewed as great opportunities to buy a high quality set of assets. The long dated nature of the majority of TLC's concessions gives us a belief that renewal is highly likely across all its concessions over time, meaning that we see TLC as (at the 50th percentile of outcomes) a growing annuity style investment.

In this report we will outline our approach to measuring earnings for TLC, as well as measuring expected 50th percentile outcomes for its games on a forecasting basis. We generally use Powerball as the mechanism for explanations, with most examples based on data prior to the most recent game change to ensure a consistent and decent sample size. Given Powerball is the largest swing factor in earnings for the group we also thought that exploring Powerball was the most relevant analysis to focus on.

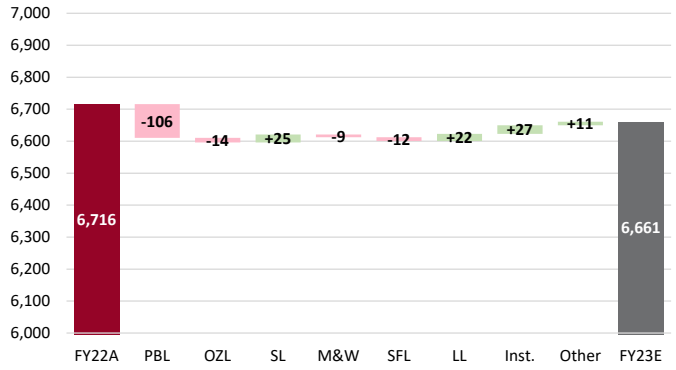
The charts below show that generally getting a Powerball forecast right or wrong has been by far the biggest swing factor in accuracy of estimating changes in TLC revenues from year to year.

Exhibit 1 - FY21-FY22 TLC Turnover waterfall - game impacts (A$m)



Source: Company Data, E&P estimates. Note: Excludes WA

Exhibit 2 - FY22-FY23E TLC Turnover waterfall - game impacts (A$m)



Source: Company Data, E&P estimates. Note: Excludes WA

This report will therefore feature 3 broad topics of discussion prior to getting to our forecasts and valuation:

- Longer term lottery revenue growth
- Estimating Entries
- Overview of our Monte Carlo simulation work

Longer term lottery revenue growth

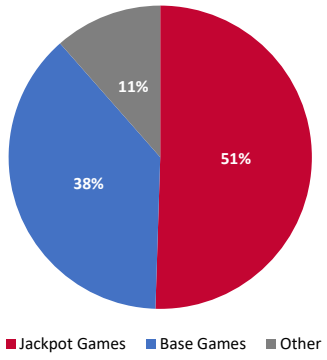
Games Portfolio

TLC's portfolio of games can be classified into two main buckets; jackpot games and base games.

1. Jackpot Games. Have a division-one prize that if not won in a given week, will continue to rise until it is won. This includes Powerball & Oz Lotto.
2. Base Games. Don't have escalating division-one prizes. Jackpots can vary but usually in a semi-consistent manner. These include Saturday Lotto, Monday Lotto, Wednesday Lotto & Set for Life.
3. Other Games. Doesn't fall into either category or are too small. This includes Instants, Lucky Lotteries and other small games (e.g. Super66/Strike). Lucky Lotteries does have elements of a jackpot game, but the draw frequency is not weekly.

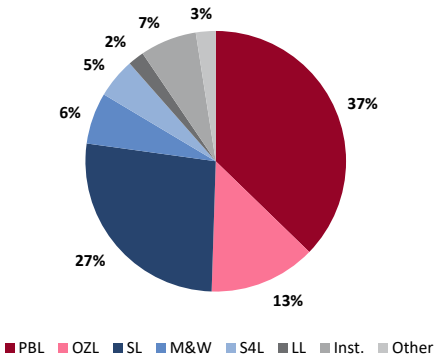
Jackpot games have observed the most growth over recent years now account for 51% of total sales. This goes further to our work being focussed on understanding Powerball the most, as it is the biggest swing factor Jackpot game.

Exhibit 3 - FY23E Lotteries Sales by Game Style



Source: Company Data, E&P Estimates

Exhibit 4 - FY23E Lotteries Sales by Game



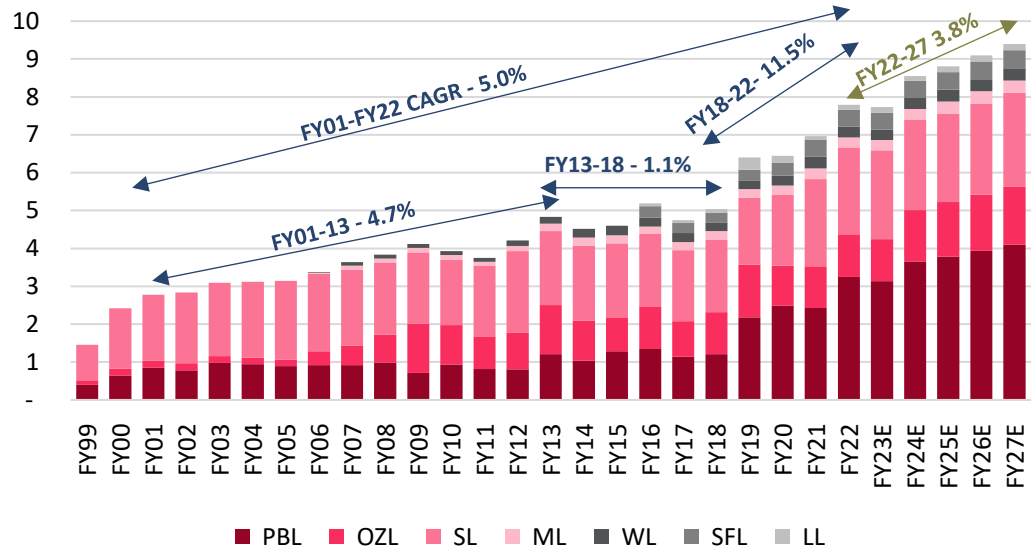
Source: Company Data, E&P Estimates

Long Run Performance

Lottery sales have grown at a healthy 5.0% CAGR since FY01. This growth has largely been aided by the FY18-FY22 period which saw 11.5% growth. We consider this period to be the digital adoption & modern Powerball era. We forecast the FY22-FY27 period to have 3.8% growth, marginally below the long-run growth rate given the strong period we just observed.

The portfolio mix has changed significantly since its origins. Powerball and Oz Lotto once battled it out for the title of most dominant Jackpot game. However, post the Apr-18 Powerball game change it has become differentiated. Powerball also became the largest game overall, which in conjunction with the new games added, reduced the reliance on the heritage game Saturday Lotto.

Exhibit 5 - Australian lotteries sales (A$bn)



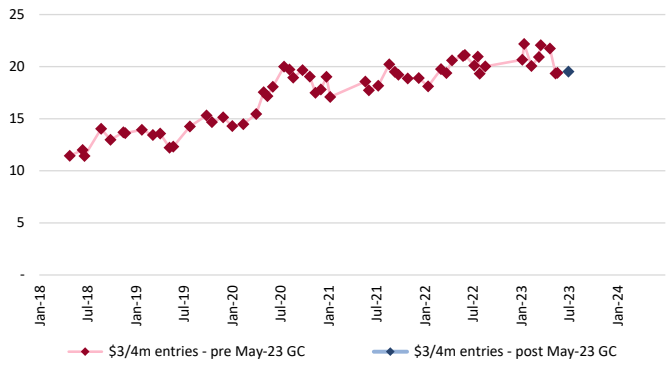
Source: Source: Company Data, E&P Estimates. Note: Includes WA & commissions. Excludes Instants & other minor games.

Like-for-Like Jackpot Entries

We believe Like-for-like entries for a given jackpot level provide a good insight into underlying growth in the lotteries game. There is both a more significant sample size to compare against for lower-level jackpots, and more frequent occurrence which can mitigate the potential impacts of things like seasonality or other unpredictable factors.

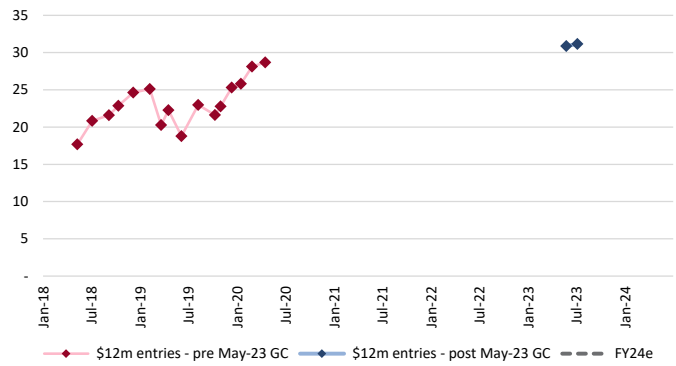
Powerball also appears to have been largely unaffected by the Oz Lotto Game change in May 2022. However, we note that despite the theoretical benefits of the game change, they have yet to materialize in better jackpot sequences and therefore more revenue. Consequently, we believe LFL entries remain marginally elevated due to the absence of cannibalisation (more in a later section) that may arrive if Oz Lotto performs more in line with its 50th percentile expectations. Below are charts outlining the growth in like-for-like jackpot entries. All the charts in general indicate a mid-single-digits growth rate or thereabouts, with the exception of the $12m and $100m for which there aren't enough jackpots consistently to be conclusive.

Exhibit 6 - PBL Entries (m's) when Jackpot is $3/4m



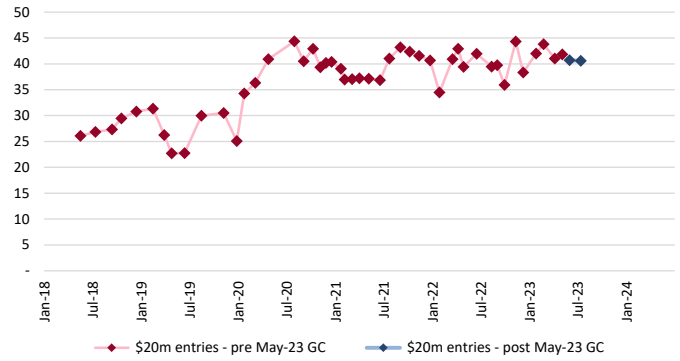
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 7 - PBL Entries (m's) when Jackpot is $12m



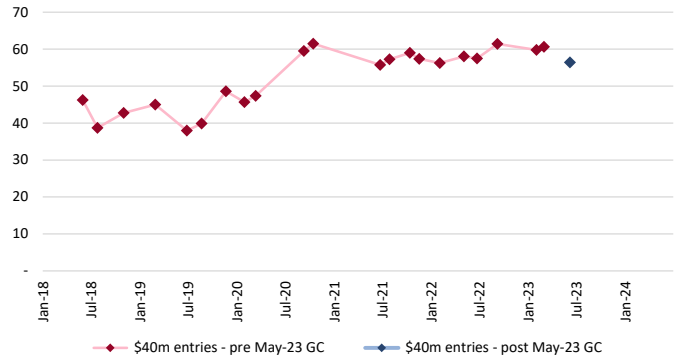
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 8 - PBL Entries (m's) when Jackpot is $20m



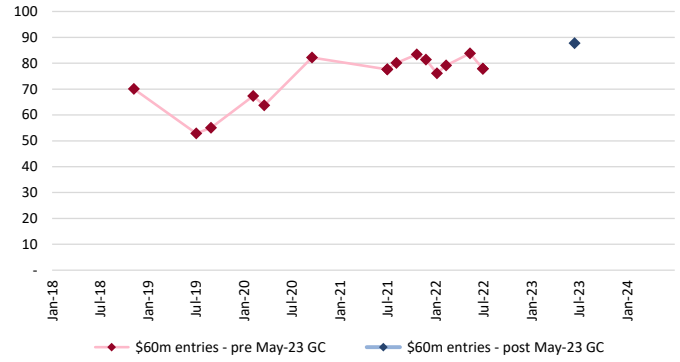
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 9 - PBL Entries (m's) when Jackpot is $40m



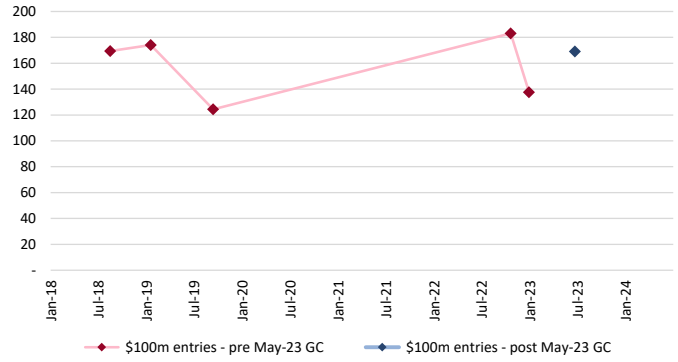
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 10 - PBL Entries (m's) when Jackpot is $60m



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 11 - PBL Entries (m's) when Jackpot is $100m

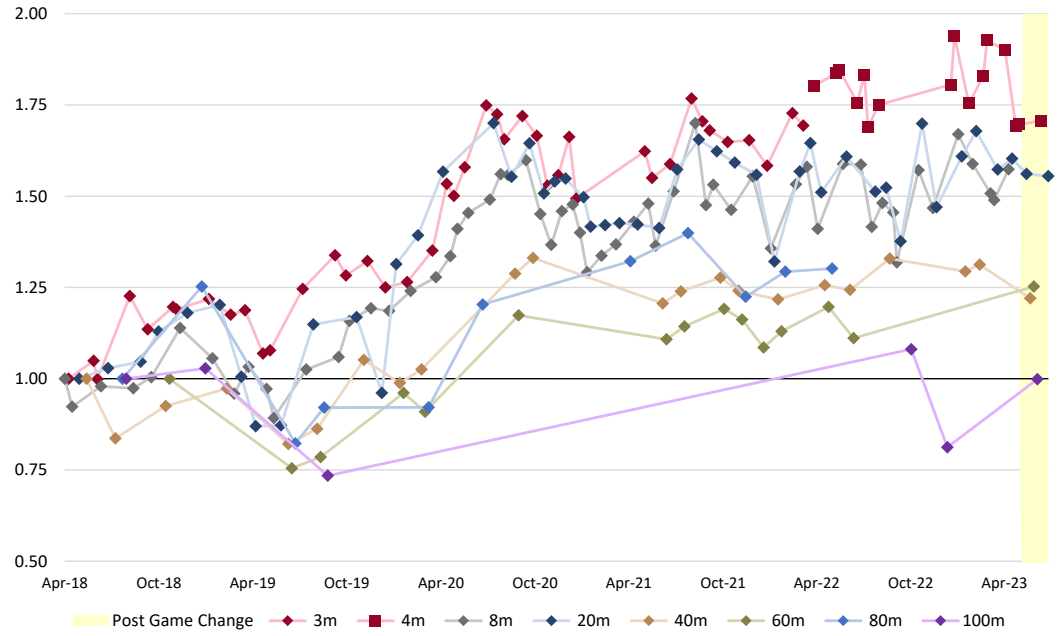


Source: Company Data, E&P Estimates. Note: Includes WA

If we compare the frequent jackpots indexed to the April 2018 game change, we can observe some common trends:

1. Most LFL jackpot entries held flat initially but then declined in 2019. This could be explained by an increase in price and lengthening of the Div-1 odds.
2. The smaller jackpots gained the most through lockdowns. We suspect this is partly due to digital adoption, which in turn leads to more players who are subscribers that play at any given jackpot level.
3. The $100m jackpots appears to have gained the least albeit on low count of jackpots to compare. We suspect this is due to two reasons: (i) digital migration allowing players greater opportunity to play large jackpots overseas such USA Mega Lotto (currently $1.3bn) via international online retailers - albeit we believe this is a smaller factor; and (ii) clustering of $100m+ jackpots which we believe may be leading to what we term 'jackpot fatigue' (more on each of these in later sections).

Exhibit 12 - Powerball like-for-like Jackpot Entries index post FY18 Game Change (Apr-18 = 1)



Source: Company Data, E&P Estimates. Note: Includes WA

Estimating Entries

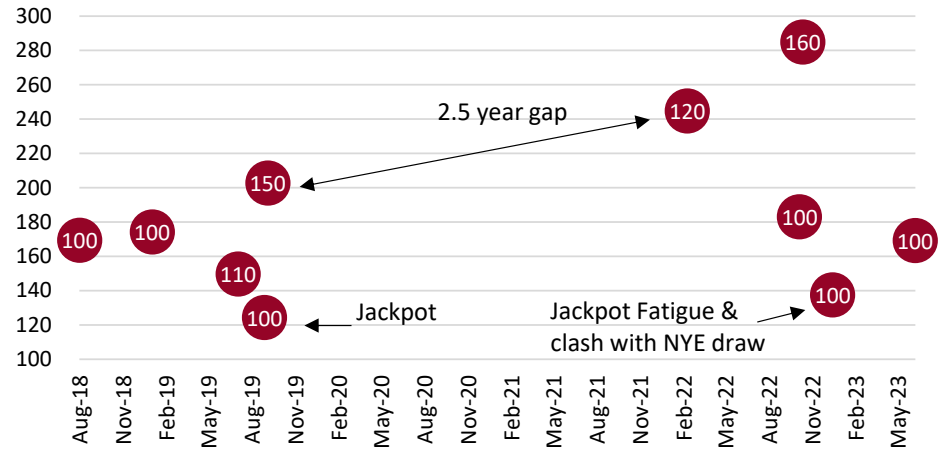
We believe that the line of best fit to model how entries work for a given jackpot is via an S-curve, that has a minimum asymptote defined by the number of weekly subscriptions, and an upper bound asymptote that is likely defined by some proportion of Australian weekly earnings. For now we don't believe jackpot sequences have challenged the upper bound of the lottery entries S-curve in Australia.

$100m draws we believe currently demarcate the steeper part of the S-curve for entries volumes

The $100m+ jackpot draw remains a significant phenomenon in Australia. There have only been twelve instances in domestic history - ten from Powerball and two from Oz Lotto. All ten instances of Powerball have come since the very successful April 2018 game change - not a coincidence given the 75.2% lengthening of the Div-1 odds.

There were five instances in quick succession post game-change, largely a function of favorable draws that were far more frequent than the theoretical average. Powerball then observed an unfavorable 2.5 year period with six consecutive $80m jackpots being won and zero $100m+ draws. The most recent group of five was aided by the acceleration of the jackpot sequence and the scrapping of 80m jackpots (more on both in later sections). The chart below shows how entries have performed with different $100m+ jackpots.

Exhibit 13 - Powerball Entries when Jackpot is 100m+ (no. mlns, LHS; jackpot size A$m, within circle)

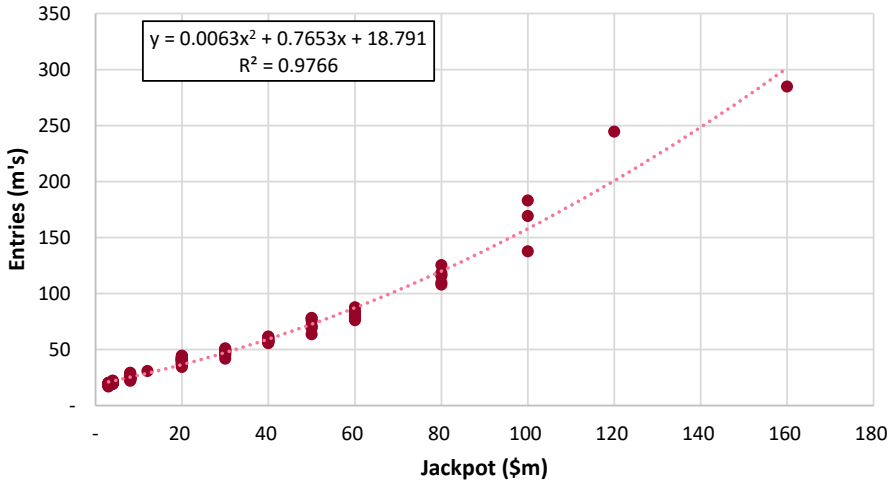


Source: Company Data, E&P Estimates. Note: Includes WA

Entries in 'normal' draws and $100m+ draws

The division-one jackpot and number of entries have demonstrated a strong positively correlated relationship over time. Across all jackpots, this relationship is best represented by a polynomial trendline (ie. not a straight line, but with some acceleration in entries as the jackpot grows). In terms of mapping out a line of best fit, for draws up to $80m there has been relatively minor variance from draw to draw, however variance increases once the draws get into the $100m+ range.

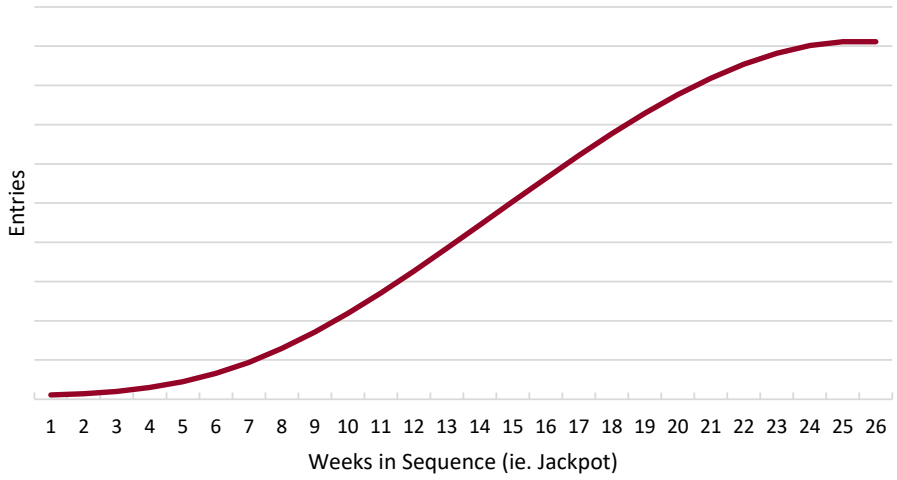
Exhibit 14 - Exponential Relationship between Powerball Entries & Jackpots.



Source: Company Data, E&P Estimates. Note: Includes WA. Data is from FY21 to pre May-23 game change

We believe the ultimate shape of the entries' volume curve resembles that of an S-curve, with an upper bound asymptote defined as some percentage of Australian disposable income or weekly income. While an incrementally climbing jackpot sequence allows players to become invested in a sequence, it also requires players to spend a larger amount of disposable income within a short time span to keep participating. The consequence of our assumption is that say a $500m draw and a $1b draw would have very similar entries, whereas we would expect entries to keep climbing relatively consistently for draws less than $500m (take $500m as an arbitrary number, not a calculated number).

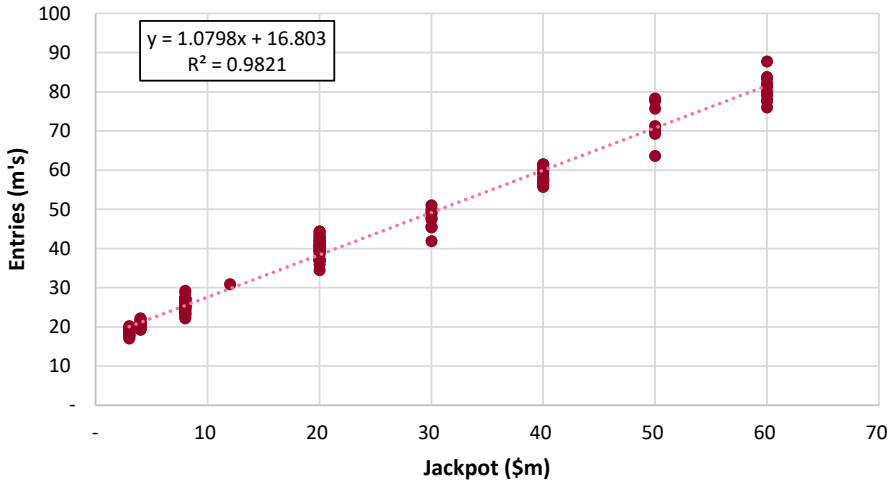
Exhibit 15 - Powerball entries - we beleive demand has an S-curve shape to it as the Jackpot sequence goes on



Source: E&P Estimates. Note: Legislation requires the maximum sequence length to be 26 weeks.

For jackpot draws that are sub $60m, a linear relationship offers the most explanatory line of best fit to historical entries data. In the chart below the 'constant' or y-axis intercept is 16.803m - we believe this gives a good approximation to the number of entries bought each week by people on recurring subscriptions or who go down to the newsagent each week and buy their numbers for the week in relation to Powerball.

Exhibit 16 - Linear Relationship between Powerball Entries & Jackpots $60m or less.

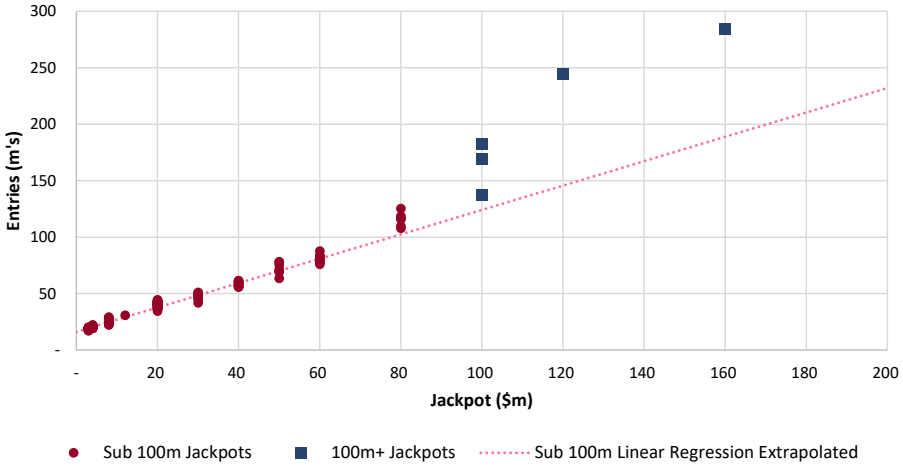


Source: Company Data, E&P Estimates. Note: Includes WA. Data is from FY21 to pre May-23 game change

$100m+ Draws stop fitting to a linear relationship

If we extrapolate the above trendline that was linear for the 'up to $60m draws' chart, we can observe the premium in entries that $100m+ draws can obtain above the linear estimation.

Exhibit 17 - Sub 100m Entries Linear Regression Exprapolation



Source: Company Data, E&P Estimates. Note: Includes WA. Data is from FY21 to pre May-23 game change

In all ten instances of $100m+ jackpots, the number of entries recorded a premium over the extrapolated linear estimation for the relevant FY. However, the premiums demonstrate significant variation. We suspect that there are four primary reasons:

1. Date in history when the draw happened (i.e. time series).
2. The number of days since the last $100m+ jackpot (ex. draws in the same sequence) which we labeled above as jackpot fatigue.
3. Whether the draw was the largest in history at the time (marketing boost).

4. Cannibalisation from other major games. In this instance we have looked at the next Saturday Lotto draw (2nd largest game) two days later & the prior Oz Lotto draw (3rd largest game and most similar) two days earlier.

The table below outlines historical $100m+ draws, entries, and then benchmarks against these four factors, all of which we believe have explanatory power.

Exhibit 18 - Key metrics that impact the Premiums from $100m+ Powerball Jackpots

Timeframe		Result				Factors			
FY	Date	Jackpot	Actual Entries	Linear Estimation (a)	Premium	Days since 100m+ draw (b)	Largest in History (at time)	Next SL Jackpot	Prior OZL Jackpot
FY19	16-Aug-18	100,000,000	169,384,236	108,613,300	56.0%	na (c)	Yes	4,359,646	2,000,000
FY19	17-Jan-19	100,000,000	174,145,592	108,613,300	60.3%	154	No	4,118,836	10,000,000
FY20	18-Jul-19	110,000,000	149,659,729	102,276,000	46.3%	182	No	4,316,189	10,000,000
FY20	12-Sep-19	100,000,000	124,361,859	94,361,000	31.8%	56	No	3,841,185	5,000,000
FY20	19-Sep-19	150,000,000	202,664,967	133,936,000	51.3%	63	Yes	4,164,796	10,000,000
FY22	24-Feb-22	120,000,000	244,631,645	153,613,000	59.3%	889	No	10,000,000	2,000,000
FY23	20-Oct-22	100,000,000	183,084,473	131,034,000	39.7%	238	No	5,024,463	5,000,000
FY23	27-Oct-22	160,000,000	284,837,414	199,770,000	42.6%	245	Yes	20,000,000	10,000,000
FY23	29-Dec-22	100,000,000	137,554,844	131,034,000	5.0%	63	No	40,000,000	10,000,000
FY23	22-Jun-23	100,000,000	169,147,583	131,034,000	29.1%	175	No	30,000,000	3,000,000

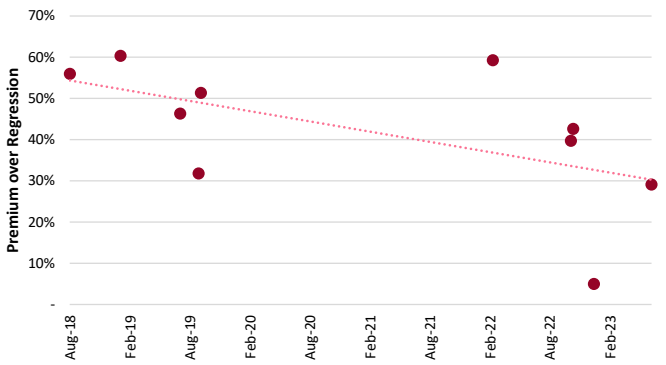
a) linear regression estimates are for sub $60m jackpots within the relevant FY
b) excludes $100m+ jackpots in the same sequence
c) first $100m+ jackpot in Powerball history

Source: Company Data, E&P Estimates. Note: Includes WA

While the dataset is somewhat limited, we believe that the premiums have been declining over time. We believe the $100m + draw is no longer viewed with quite the same mystique it once had, and also at the margin that large international jackpots via online operators offer an alternate product that some people in Australia are participating in.

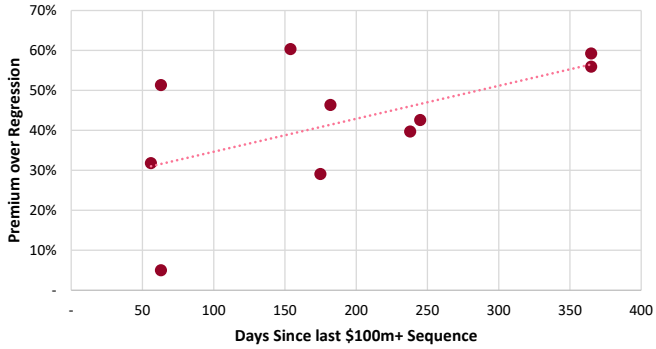
We believe that cannibalisation and jackpot fatigue are factors that impact entries. The real world dynamics we believe impact here are that at a physical bank account level if a player has spent a large amount on a large jackpot recently, their funds to spend again are more likely to be depleted. Secondly, the marketing impact of a $100m+ draw has a certain impact if it hasn't happened for a while, but can become less interesting if the consumer has recently been exposed to the same marketing messages.

Exhibit 19 - $100m+ Powerball Jackpot Premium over Time



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 20 - $100m+ Powerball Jackpot Premium based on the number of days since most recent $100+m Jackpot



Source: Company Data, E&P Estimates. Note: Includes WA. Note: Last $100m + jackpot does not include jackpots in the same sequence. Days are capped at 365.

Cannibalisation and fatigue

We believe cannibalisation and fatigue (different words for the same single impact) is a factor in overall entries, with the two lowest premiums (entries for a $100m+ draw vs linear extrapolation) occurring two days before a $30/40m Saturday Lotto Megadraw. This is a limited sample size, as such it is hard to prove out mathematically, however the concept feels intuitively comfortable.

In the table and charts below, we have tracked the performance of frequent Powerball jackpots vs. the full year average (i.e. premium) against the size of the surrounding jackpots for Saturday Lotto and Oz Lotto.

Under this approach, the expectation for fatigue/cannibalisation is that when a Powerball draw is near a large Oz Lotto jackpot (on the Tuesday prior to Powerball) then the Powerball draw should experience below average entries, while if the Powerball draw is following a small Oz Lotto jackpot then the draw should have above average entries.

In the 43 instances where the last Oz Lotto jackpot two days prior was $2m, the average Powerball jackpot was +2.2% above its full year average & was greater than zero 67.4% of the time. This is a considerably superior performance to the five instances where the last Oz Lotto jackpot was $50M, yielding an average -9.3% underperformance vs the full year average number of entries, and with no instances above zero. The same relationship can be observed when analyzing the next Saturday lotto draw, two days after the Powerball draw, as shown in the table and charts below.

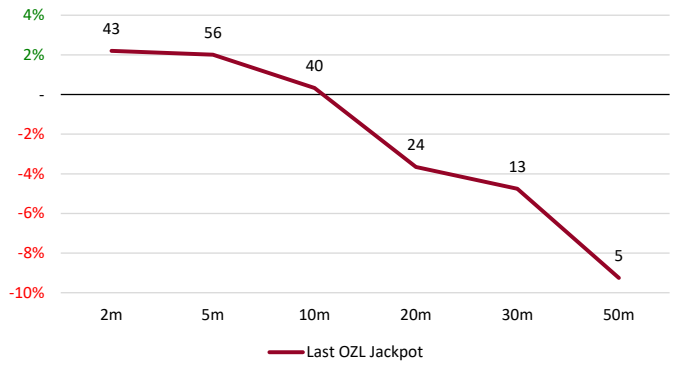
Exhibit 21 - Cannibalisation of Powerball based on surrounding Oz Lotto & Saturday Lotto Jackpots

OZL jackpot	Count		Average PBL premium		% PBL premium >0	
	Last	Next	Last	Next	Last	Next
2,000,000	43	50	2.2%	0.1%	67.4%	58.0%
5,000,000	56	47	2.0%	1.7%	66.1%	66.0%
10,000,000	40	47	0.3%	2.2%	47.5%	66.0%
20,000,000	23	19	-3.8%	-1.1%	30.4%	36.8%
30,000,000	13	15	-4.8%	-3.7%	23.1%	13.3%
50,000,000	5	5	-9.3%	-7.0%	0.0%	0.0%

SL Jackpot	Count		Average PBL premium		% PBL premium >0	
	Last	Next	Last	Next	Last	Next
$3m-$10m Base Draws	175	179	-0.1%	0.4%	51.4%	55.3%
$20m-$22m Superdraw	38	36	0.5%	-1.4%	55.3%	41.7%
$30m-$40m Megadraw	6	3	-3.8%	-9.2%	33.3%	33.3%

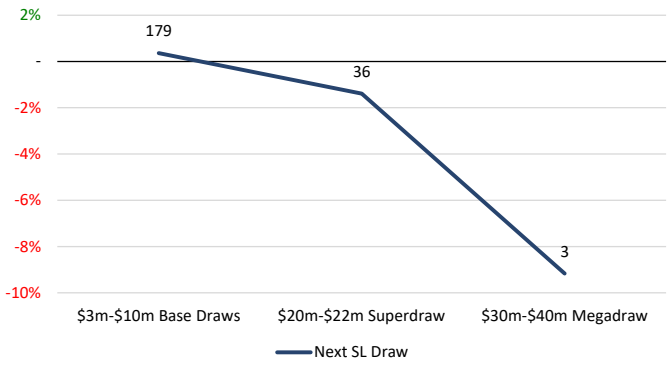
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 22 - Like-for-Like Jackpot Powerball sales vs. FY average based on the last Oz Lotto Jackpot (# of instances as label)



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 23 - Like-for-Like Jackpot Powerball sales vs. FY average based on the next Saturday Lotto Jackpot (# of instances as label)



Source: Company Data, E&P Estimates. Note: Includes WA

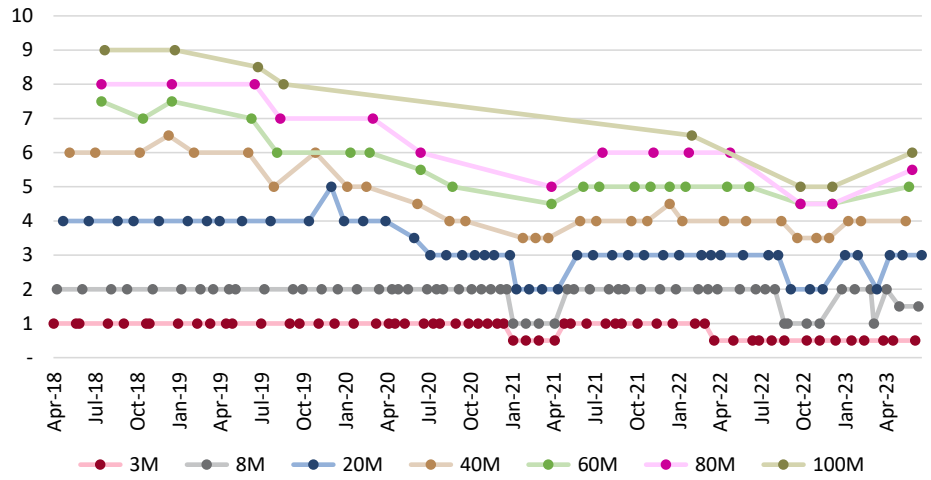
Powerball game mechanics important to modelling

Accelerated Powerball Sequence

Since the April 2018 game change, Powerball has incrementally adopted faster sequences. For example, the current sequence reaches the $40m jackpot (brown line) in four weeks, compared to the six weeks it took back in FY19. While management has full discretion to employ whichever sequence they deem most suitable, the speed-up is primarily a function of increased LFL jackpot entries. This forces sequence acceleration in order to ensure the prize pool payout meets the minimum requirements outlined in the legislation (explored in the next section).

The chart below shows how the Powerball sequence has shortened over time, with larger jackpots being reached in less weeks as time goes on.

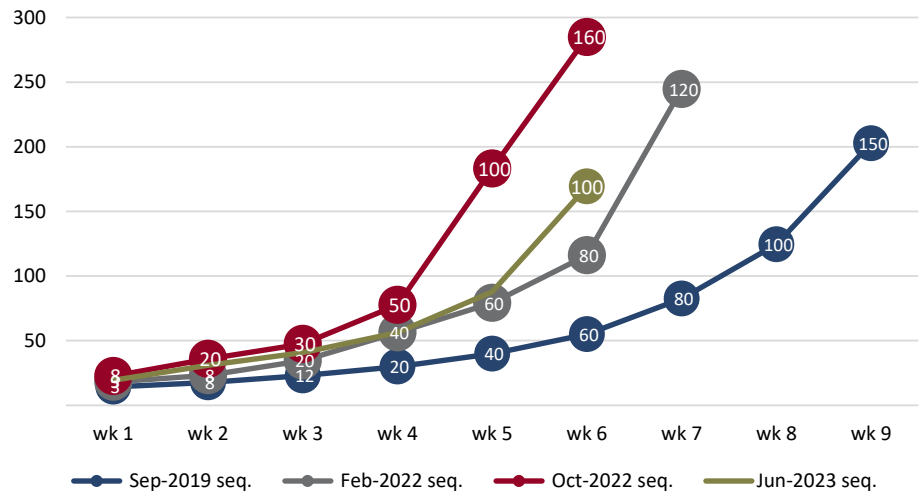
Exhibit 24 - Weeks to reach a given Jackpot size since the Apr-18 Powerball game change.



Source: Company Data, E&P Estimates. Note: (1) Includes WA. (2) .5 weeks implies jackpot was above greater than the benchmark (Left chart). For example, when the wk. 6 jackpot was $80m and wk.7 jackpot $120m, the $100m jackpot is considered reached in 6.5 weeks.

As previously mentioned, this is part of the reason we observe $100m+ draws more frequently. The initial $150m draw in September 2019 took nine weeks to reach, while the record $160m draw took only six. The chart below illustrates this speed-up towards $100m+ jackpots as entries have risen and the jackpot sequence has shifted faster.

Exhibit 25 - $100m+ Powerball Jackpot Sequences (entries no. mlns, LHS; jackpot size A$m, within circle)



Source: Company Data, E&P Estimates. Note: Includes WA

Since FY21, like-for-like jackpot entries have remained relatively consistent. Consequently, the sequences have been stable as well. The 'normal' sequence (4,8,20,40,60,80,120) and the 'accelerated' sequence (8,20,30,50,100,160) were used in approximately an 80/20 split pre the May-23 game change. Given the prize payout dynamic which we discuss in the next section, this has meant that the sequence has needed to fluctuate between a standard and accelerated sequence (and in certain instances a slower sequence) in order to maintain an appropriate payout ratio.

The accelerated sequence achieves larger jackpots faster and therefore more sales. We estimate this to be ~24.2% over a full-year simulation. The downside of the accelerated sequence is that it has a higher payout ratio and is therefore unsustainable in its current format vs the regulatory payout ratios, particularly once taxation is taken into account. In order for the accelerated sequence to be sustainable stand-alone (ie. become the normal sequence) you would need some combination of the following: (i) LFL jackpot entries increase, (ii) the subscription price increases, (iii) increases in division-1 prize allocation, or (iv) legislation changes to payout ratios.

Prize Payout Ratios and the Prize Reserve Fund

The Powerball game rules legislate that the overall return to player must not be less than 60% of the total player subscriptions. This is straightforward in principle but requires a deeper understanding to apply in practice. Firstly, prize allocations are specifically based on 'subscriptions' and not 'sales'. The subscription price is the authorized cost of entry whilst the sales price includes the retail commission. Pre May 2023 game change, the Powerball subscription price was $1.10 and the standard ticket cost is $1.21. The recent 200bps increase in retailer commission that accompanied the game change does not impact the prize payout dynamic. However, the 10c increase in subscription price will have a major impact.

The second key element to understand is that the 60% payout requirement is over the long term, not each individual draw or sequence. The rules allow for a prize reserve fund (PRF) that offers TLC a level of flexibility to 'promise' a specific division-one prize before the draw. Without this structure, the division-one prize would not be known until after the draw. Furthermore, the PRF allows TLC to vary the promised division-one jackpot such that they could be considered 'rounded' or 'clean' prizes as well as to manage short-term discretionary impacts. The current split of the 60% overall payout is a 57.5% definitive return to player and a 2.5% prize reserve fund over a full sequence. This has varied through time and varies by game. In practice, this means that there is a hard floor (57.5%) in the proportion of prize payout for a sequence, but there is no ceiling.

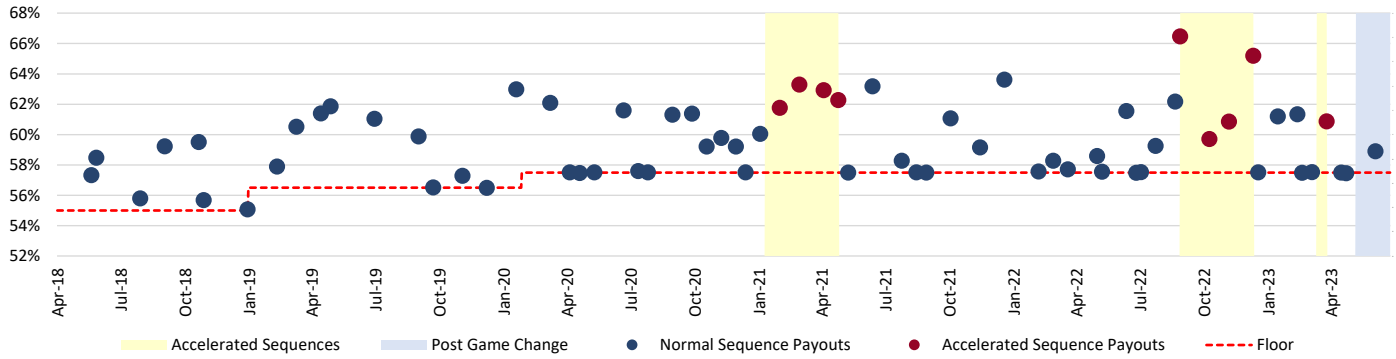
The final key point is that the 60% prize allocation is split across all nine divisions. The important distinction is between division-one prizes and non-division-one prizes. Every week, irrespective of whether there is a division-one winner, divisions two to-nine will receive a consistent proportion of that week's subscriptions. The division-one prize allocation will vary as a proportion

of the overall sequence subscriptions, but long-term will be made up of both the allocation towards division one (35.0% of the 57.5% minimum payout pre game change, 33.5% post game change) and the PRF allocation (2.5% of subscriptions).

Tracking the Sequence Payout

Historically we have observed the sequences to have some variation in payout ratio (return to player), with the average being around 60% as expected and legislated.

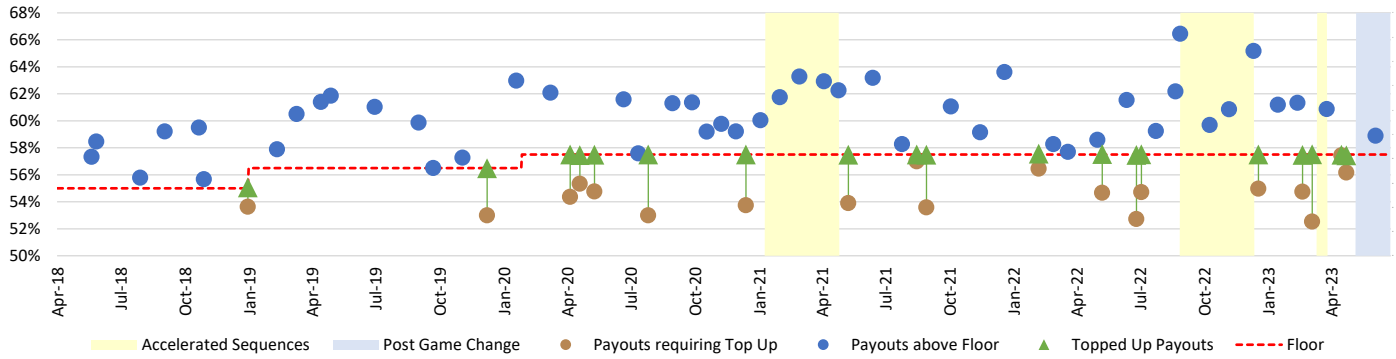
Exhibit 26 - Powerball Sequence Payout Ratios



Source: Company Data, E&P Estimates

The graph above demonstrates that the overall payout for any given sequence will be no less than the floor requirement at the time. However, should the promised division-one prize not be sufficient to meet this requirement, TLC will 'top up' the division- one prize to meet legislation. This concept in practice can be seen below.

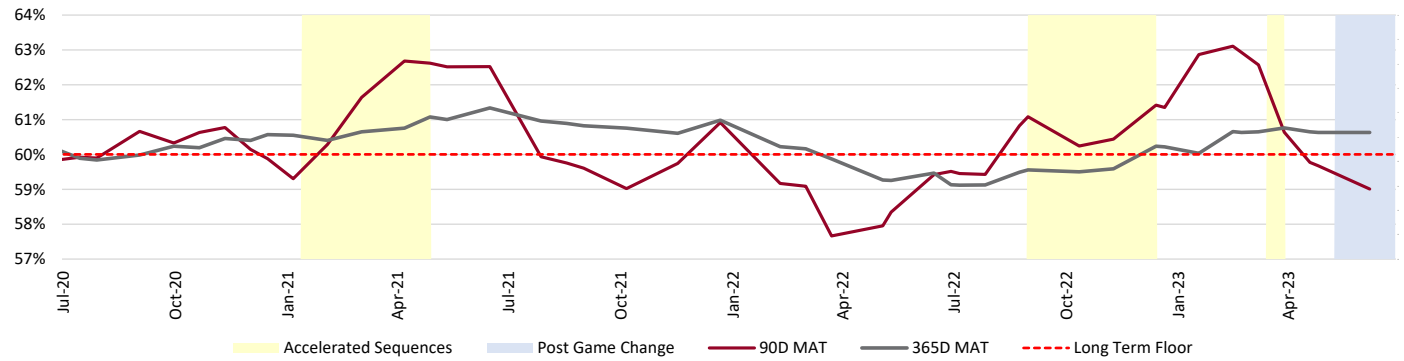
Exhibit 27 - Demonstration of how Div-1 Prizes get "Top Ups" to meet the Game Regulated Sequence Prize Floor



Source: Company Data, E&P Estimates

Over time, we have typically observed the moving average total (MAT) overall payout ratio to become elevated during and immediately after the accelerated sequences.

Exhibit 28 - Powerball Sequence Payout Ratio Moving Average Total (MAT)



Source: Company Data, E&P Estimates

This is an expected outcome as it is the reason for acceleration typically. While there will be some variation for the size of the jackpot, accelerated sequence payouts ratios will almost always be higher at the LFL jackpot level as the purpose of acceleration is (typically) to deplete the overfunded prize reserve fund. For example, if we compare the $8m jackpot in the normal sequence (week-2) against the accelerated sequence (week-1), then the normal sequence has an extra $4m jackpot (that doesn't go off) to accumulate funds for the div-1 prize.

The ideal low-end jackpots maximise for entries while paying out between 57.5% - 60%

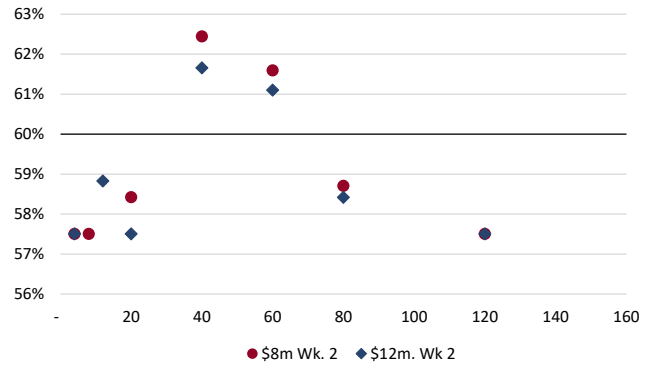
Finding an ideal jackpot sequence is challenging given the sensitivity of the payout ratios. The long-run Powerball allocation towards div-1 is 21.7625% of subscriptions, with a floor of 20.1250% for any particular sequence, offering minimal wiggle room. Management has also shown a preference to opt for consistency where possible, even when accelerating the sequence. For example, the accelerated sequence maintains the traditional $8m & $20m jackpots but skips the beginning $4m, rather than increasing the original normal sequence jackpot size by a certain percentage (e.g. $5m, $10m, $25m).

We believe it is unfavorable to have a theoretical payout ratio (based on the advertised jackpot) below the floor 57.5% level. This is because we suspect that you are indirectly leaving money on the table as entries and therefore revenue has a strong positive correlation to advertised jackpot, and then regulations force TLC back to paying at least 57.5% on a sequence anyway. Most players are unaware of the statistics though and just aware of the advertised jackpot.

We have observed this with the $8m jackpot during the normal sequence (week-2). The $8m jackpot has gone off six times since FY21, and each time the division-one prize has been topped up by an average of $2.15m or 27%. Had TLC promised this amount to begin with ($10.15m jackpot, or rounded $10m), they would have likely earned more entries without having to pay any extra division-one prize (note they still would have had to pay an increased amount of non-division-one prizes if they do in fact increase subscriptions). The only downside is that more entries increase the probability of the jackpot going off. But this is not enough downside to offset the gains from additional subscriptions.

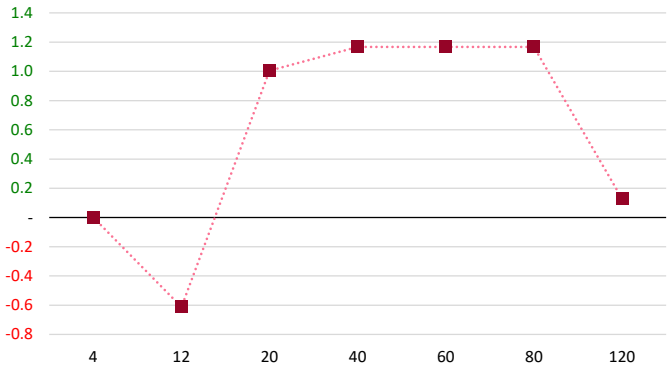
With the recent game change, we believe management has identified this and made the right adjustment. Their sequence now has $12m as the week-2 jackpot and this has theoretical benefits for both pre & post game change. Pre game change, we suspect that this would increase entries by ~4.7m by moving to the $12m jackpot (post game change there is not a lot of data yet). The charts below show the impact of moving the old game from a $8m week 2 to a $12m week 2 in terms of payout ratios (left side), and in terms of net impact on PRF (right side, generally positive which enables greater jackpots and therefore likely revenues).

Exhibit 29 - Change in Seqeunce Payout Ratio by shifting to $12m jackpot in Wk.2 - pre game change



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 30 - Increase in Net Sequence Prize Reserve Fund Accumulation by shifting to $12m jackpot in Wk.2 - pre game change



Source: Company Data, E&P Estimates. Note: Includes WA

The table below puts some of this into raw numbers which may be easier to digest. It contains the following data points:

- On the left hand side we show the advertised jackpot, and the expected entries and $ subscription value (under old game).
- In the second section "subscription allocation" we show the return to game (40%), non-Div1 prizes paid and then the theoretical Div1 prize allocation for that draw stand-alone.
- The third section shows the movement in the Prize Reserve Fund first on a stand-alone draw basis, and secondly on a cumulative basis. What can be seen in this section is that on average under the old Powerball game a sequence that paid out the Jackpot during a $40m or $60m draw would typically deplete the Prize Reserve Fund, whereas sequences triggered above or below those draws would typically build the Prize Reserve Fund.
- Finally we show the amount needed to be paid to Div1 in order to hit either a minimum 57.5% payout ratio, or to hit a 60% through-time legislated payout ratio.

Exhibit 31 - Powerball example workings for the Prize Reserve Fund of the Normal Sequence - pre game change

Jackpot & Avg. Entries			Subscriptions Allocation			PRF Accumulation if Jackpot		J/Pot to reach breakeven	
Jackpot	Entries	Subscriptions	TLC Take (40%, pre Taxes)	Non Div-1 Prizes	D1 Prize Allocation	Week Only	Cumulative Sequence	57.5%	60.0%
4,000,000	20,636,433	22,700,077	9,080,031	8,484,154	5,135,892	1,135,892	1,135,892	4,568,390	5,135,892
8,000,000	25,531,802	28,084,982	11,233,993	10,496,762	6,354,227	-1,645,773	3,490,120	10,220,493	11,490,120
20,000,000	40,217,720	44,239,492	17,695,797	16,534,510	10,009,185	-9,990,815	1,499,305	19,123,691	21,499,305
40,000,000	58,672,856	64,540,142	25,816,057	24,121,878	14,602,207	-25,397,793	-3,898,488	32,112,394	36,101,512
60,000,000	80,174,223	88,191,645	35,276,658	32,961,627	19,953,360	-40,046,640	-3,945,129	49,860,963	56,054,871
80,000,000	115,713,692	127,285,061	50,914,024	47,572,792	28,798,245	-51,201,755	4,853,116	75,477,081	84,853,116
120,000,000	244,631,645	269,094,809	107,637,924	100,574,185	60,882,701	-59,117,299	25,735,817	129,632,412	145,735,817

Source: Company Data, E&P Estimates. Note: includes WA. Prize allocations don't include top ups to reach sequence floor.

Recent Powerball Game Change

May 2023 Updates

The May-23 Powerball game change is very minor relative to the previous changes made to Powerball (Apr-18) & Oz Lotto (May-22). While both previous changes included a price rise, the most significant impact was the lengthening of the division-one odds, which the May-23 change does not have. The lengthening of the odds decreases the probability of the jackpot going

off, and consequently increases the theoretical number of large jackpots on offer. However, despite the minor price increase, we expect this change to be positive.

Changes to ticket prices have historically been greater than the corresponding decline in entries (i.e. some inelasticity of demand). Further, the increase is smaller than the April 2018 change, and could almost be considered in-line with inflation, reducing the likelihood of negative customer sentiment. The question then becomes whether players purchase 10 tickets or $10 worth of tickets. We suspect the majority of players lean toward the former given: (i) players wouldn't want to reduce their chances of winning, (ii) purchasing 9 tickets feels odd, (iii) TLC purchase options are by entries not price, and (iv) players cannot physically purchase exactly $10 worth of tickets.

Exhibit 32 - Summary of Recent Game Changes

	Powerball	Set for life	Saturday Lotto	Oz Lotto	Powerball	M&W Lotto
Date	**19-Apr-18**	**23-Mar-20**	**10-Oct-20**	**17-May-22**	**25-May-23**	**TBD**
Game changes						
Odds (Div. 1)	Longer	-	-	Longer	-	
Odds (Any prize)	Shorter	Shorter	Shorter	Shorter	-	
Additional division	Div 9	Div 2 (1)	-	-	-	
Price increase	Yes	Yes	Yes	Yes	Yes	
Subscription Price						
Old	$0.85	$0.55	$0.65	$1.20	$1.10	
New	$1.10	$0.60	$0.75	$1.30	$1.20	
Increase	29.4%	9.1%	15.4%	8.3%	9.1%	
Combinations						
Old	76,767,600	-	-	45,379,620	134,490,400	
New	134,490,400	-	-	62,891,499	134,490,400	
Increase	75.2%	-	-	38.6%	0.0%	

Source: Company Data, E&P Estimates

The Powerball game change was only a 10c increase in the subscription price. However, it was accompanied by an increase in the retailer commission, meaning the actual ticket price for the player increased by 14c (11.34%). At the LFL jackpot level, players are marginally worse off as they pay more for the same division-one jackpot (non-division-one prizes remain as a proportion of subscriptions).

Exhibit 33 - May-23 change in Powerball Ticket Price

Entry Type	Previous Price	New Price
8 game entry	$9.70	$10.80
PowerHit entry	$24.30	$26.95

Source: The Lott

The other more subtle change is a change in the prize pool allocation. The change reduced the allocation towards division-one. This in turn will increase the payout ratios, decrease the net prize fund accumulation for sequences and decrease the potential for acceleration. However, players are theoretically better off at each LFL jackpot level as non-division-one prizes increase for the same jackpot. This may help improve entries (if players are even aware), or potentially increase the amount of winnings reinvested.

Exhibit 34 - May-23 change in Powerball Prize Allocation

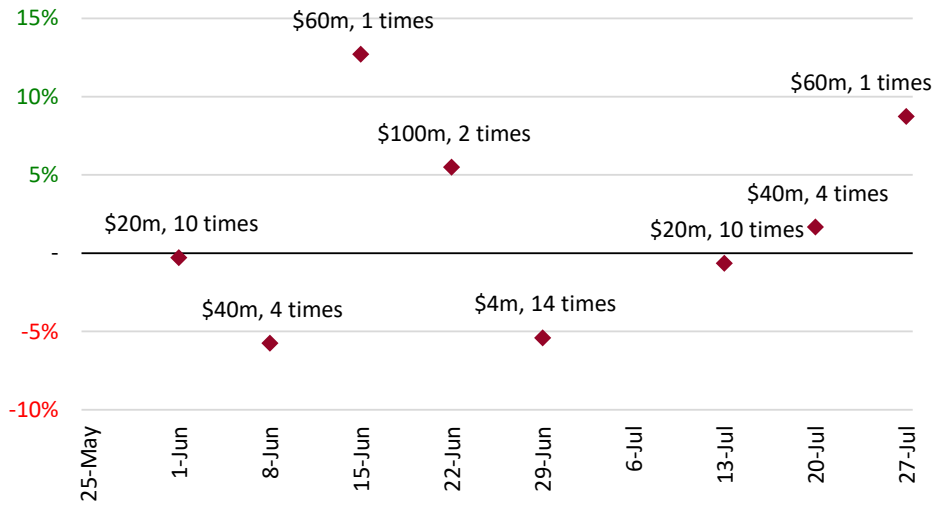
Prize Division	Previous Allocation of Prize Pool (up to and including Draw 1409)	New Allocation of Prize Pool (effective from Draw 1410)
Division 1	35.00%	33.50%
Divsion 2	1.80%	1.85%
Division 3	1.10%	2.30%
Division 4	2.00%	2.00%
Division 5	1.50%	1.50%
Division 6	9.70%	9.40%
Division 7	7.60%	7.30%
Division 8	15.00%	15.50%
Division 9	26.30%	26.65%

Source: The Lott

Changes in Like-for-Like Jackpot Entries

Although a limited sample size, we have tracked the like-for-like jackpot entries below since the May-23 game change against the 12 months pre game change. At first glance, entries appear to have improved more often than not. However, the three that considerably exceed were the $60m & $100m jackpots, both of which had few instances in the prior 12 months and may be non-representative due to low sample size. We also note the $100m draw under the new game rules compares against the particularly weak Dec-23 draw and that factors such as prior cannibalisation/fatigue or macro backdrop have not been considered.

Exhibit 35 - Change in Like-for-Like Jackpot Entries post May-23 game change vs. the 12 months pre game-change



Source: Company Data, E&P Estimates. Note: comments are like-for-like jackpot size & frequency of jackpot in the 12 months pre game change.

We expect like-for-like entries to decline by approximately -3% vs. the trailing 12 months pre game change. Although a decrease, this would still represent a ~71% retention of the increase in the ticket price. We also believe that Oz Lotto has had very unfavorable draws recently, which has marginally inflated Powerball's LFL entries. Any decline in entries that does occur has the upside of reducing the probability of a winner and increasing the theoretical probability of spending more weeks at higher jackpot levels.

Changes to Sequence

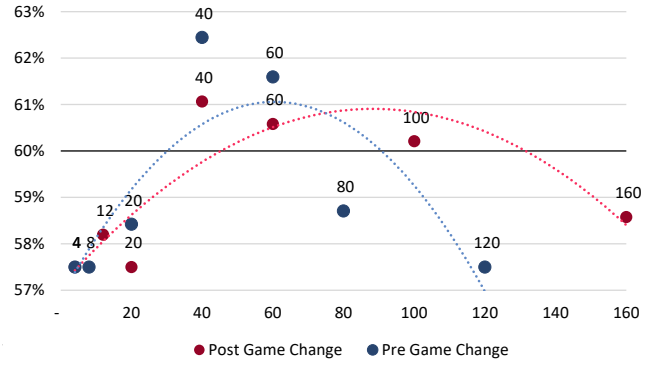
We believe management has noticed some concepts we have explored and made smart changes to the sequence. The first is changing the week-2 jackpot to $12m, which increases the sequence payout ratio above the floor. They have also scrapped the $80m jackpot and gone straight to $100m, increasing the frequency of obtaining the premium that accompanies $100m + jackpots. We are yet to see what the accelerated sequence will look like under the new game though (presuming we are seeing a new base game initially).

Changes to Prize Payouts

As previously mentioned, the increase in subscription price will impact the prize payout dynamic. Had the increased subscription price been the only change (+9.1%), then a reduction of the same proportion in LFL entries would leave the payout ratio's the same as pre game-change. However, with the division-one prize allocation declining, it would take a -4.2% drop in LFL entries to offset the subscription price increase (excl. commissions).

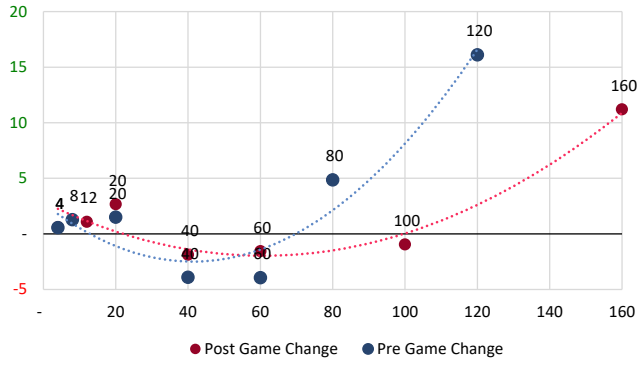
The culmination of the four updates vs. pre game change (entries, sequence, subscription price & prize allocation) result in the following payout ratios. The overall trend is largely the same, but the payouts are lower and flatter, which are both positives.

Exhibit 36 - Theoretical Payout Ratio's (%, y-axis) based on Jackpot (A$m, x-axis) - pre vs. post May-23 game change with -3% LFL Entries



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 37 - Theoretical Sequence Surplus/Defecit (A$m, y-axis) based on Jackpot (A$m, x-axis) - pre vs. post May-23 game change with -3% LFL Entries



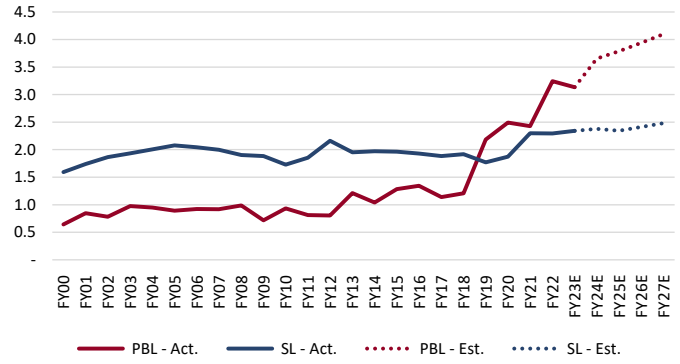
Source: Company Data, E&P Estimates. Note: Includes WA

Monte Carlo Simulation approach

Introduction to our Monte Carlo Simulator

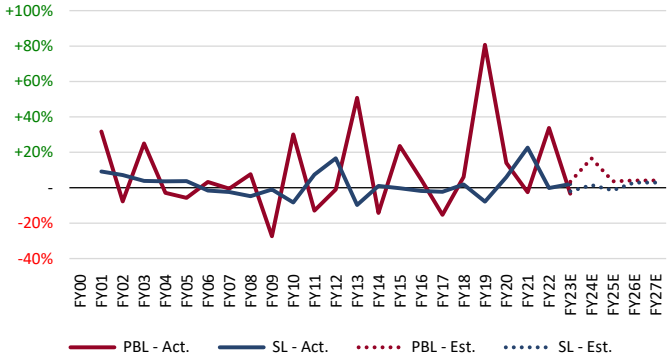
TLC is a unique company, whereby it can simply get 'lucky or unlucky' on a year-by-year basis. Consequently, any randomness of lottery draws should be considered short-term impacts, not indicators of underlying strength or weakness. Below we demonstrate a tale of two games (Powerball & Saturday Lotto) to better illustrate this premise. Since the April 2018 game change, Powerball has grown at a phenomenal rate to overtake Saturday Lotto as the biggest game by turnover. It is however, very volatile. Saturday Lotto meanwhile has not had more/less than +/-10% growth when taking out the years with game changes & covid.

Exhibit 38 - Powerball vs Saturday Lotto Sales ($bn.)



Source: Company Data, E&P Estimates

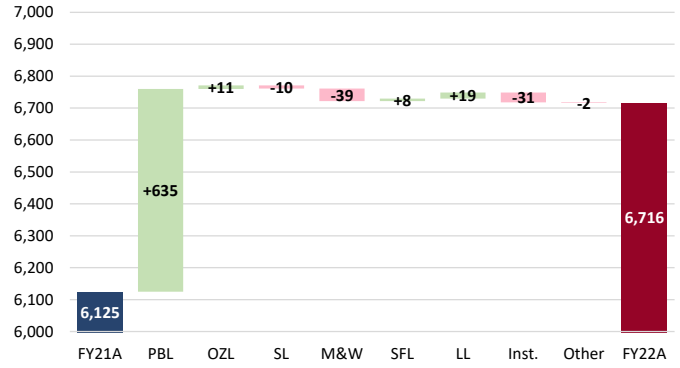
Exhibit 39 - Powerball vs Saturday Lotto Sales yoy growth (%)



Source: Company Data, E&P Estimates

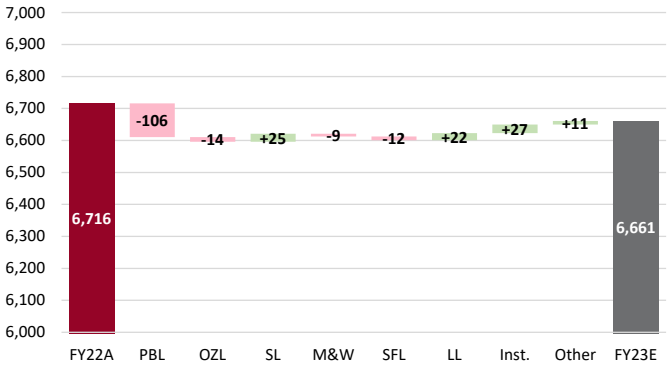
We can also see that over the past two years, movements in overall sales have primarily been driven by Powerball. This is not overly unsurprising given the randomness involved. However, we do note that Oz Lotto did not receive the theoretical benefits of the May-22 game change and that none of the other games received a significant game change that would stimulate growth.

Exhibit 40 - FY21-FY22 Lotteries sales waterfall - game impacts (A$m)



Source: Company Data, E&P estimates. Note: Excludes WA

Exhibit 41 - FY22-FY23E Lotteries sales waterfall - game impacts (A$m)



Source: Company Data, E&P estimates. Note: Excludes WA

If you were to estimate FY24 Powerball sales with an FY23 base, chances are you would be wrong, because the starting point is also wrong. This is why we estimate sales for Powerball & Oz Lotto using our Monte Carlo simulator. We estimate these two games account for 51% of sales in FY23, and forecast that they continue to take share.

Picking probabilities by Jackpot size

For an individual entry in Powerball, the probability of winning the jackpot is 1 in 134,490,400, being the number of combinations available to choose from in making an entry.

From here on out for abbreviation n = 134,490,400. From here, the probability of winning = 1/n, and the probability of not winning = 1 - 1/n for a single entry.

To then look at the probability of lots of random entries not winning, we can take the formula for a single entry and multiply it out to the power of total entries. We multiply instead of add due to the possibility that entries will choose the same numbers. This is an important distinction as the probability of jackpots occurring is based off 'unique' entries, not total entries. Therefore, our formula for 2 entries = (1-1/n) x (1-1/n) = the likelihood of the jackpot not going off with 2 random entries that have a miniscule possibility of being the exact same entry.

Probability of no jackpot with 1 entry = 99.9999992564525%

Probability of no jackpot with 2 entries = 99.999998512905%

This means an improvement in the likelihood of the jackpot triggering of +0.000000743547% (there are more digits to this but we have truncated).

The largest number of entries Powerball has seen in the current game rules is (according to our workings) 284,877,283 entries, which has a probability of ~87.98% that the jackpot will be triggered, calculated as:

Probability = 1 - (1-1/134,490,400)^284,877,283 = ~87.98%

If theoretically Powerball saw 1,000,000,000 (1 billion) entries the probability of the Jackpot going off is 99.941%.

The table below outlines our analysis of what the number of entries in each Powerball draw since FY21 meant for the probability of a given draw or sequence seeing the jackpot paid out

Exhibit 42 - Historical variance in like-for-like Jackpot Entries and the impact on probabilities

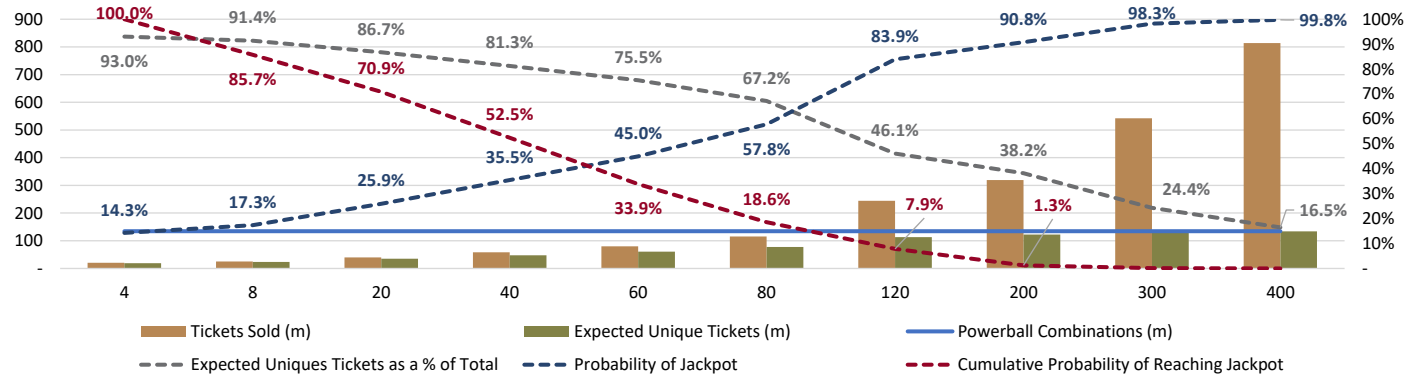
NORMAL SEQUENCE - pre GAME CHANGE												
Jackpot Frequency		Entries			Entries vs. FY avg.		Probability of Jackpot going off			Probability of reaching J/Pot		
Jackpot	Count	Min	Average	Max	Min	Max	Min	Average	Max	Min	Average	Max
4,000,000	15	19,329,129	20,636,433	22,177,022	-6.3%	7.5%	13.4%	14.2%	15.2%	100.0%	100.0%	100.0%
8,000,000	38	22,190,279	25,531,802	29,212,562	-13.1%	14.4%	15.2%	17.3%	19.5%	86.6%	85.8%	84.8%
20,000,000	31	34,469,991	40,217,720	44,346,019	-14.3%	10.3%	22.6%	25.8%	28.1%	73.4%	70.9%	68.2%
40,000,000	12	55,754,455	58,672,856	61,499,424	-5.0%	4.8%	33.9%	35.4%	36.7%	56.8%	52.6%	49.1%
60,000,000	9	76,041,502	80,174,223	83,810,205	-5.2%	4.5%	43.2%	44.9%	46.4%	37.5%	34.0%	31.1%
80,000,000	6	107,889,410	115,713,692	125,405,740	-6.8%	8.4%	55.2%	57.7%	60.6%	21.3%	18.7%	16.7%
120,000,000	1	244,631,645	244,631,645	244,631,645	-	-	83.8%	83.8%	83.8%	9.6%	7.9%	6.6%

ACCELERATED SEQUENCE - pre GAME CHANGE												
Jackpot Frequency		Entries			Entries vs. FY avg.		Probability of Jackpot going off			Probability of reaching J/Pot		
Jackpot	Count	Min	Average	Max	Min	Max	Min	Average	Max	Min	Average	Max
8,000,000	38	22,190,279	25,531,802	29,212,562	-13.1%	14.4%	15.2%	17.3%	19.5%	100.0%	100.0%	100.0%
20,000,000	31	34,469,991	40,217,720	44,346,019	-14.3%	10.3%	22.6%	25.8%	28.1%	84.8%	82.7%	80.5%
30,000,000	9	41,866,553	47,520,751	51,026,175	-11.9%	7.4%	26.8%	29.8%	31.6%	65.6%	61.3%	57.9%
50,000,000	7	63,615,310	72,275,926	78,306,877	-12.0%	8.3%	37.7%	41.6%	44.1%	48.1%	43.1%	39.6%
100,000,000	2	137,554,844	160,319,659	183,084,473	-14.2%	14.2%	64.0%	69.6%	74.4%	30.0%	25.2%	22.1%
160,000,000	1	284,837,414	284,837,414	284,837,414	-	-	88.0%	88.0%	88.0%	10.8%	7.6%	5.7%

Source: Company Data, E&P Estimates. Note: entries averages & count are irrespective of the jackpot sequence at the time.

The graph below also demonstrates the relationship between all the factors.

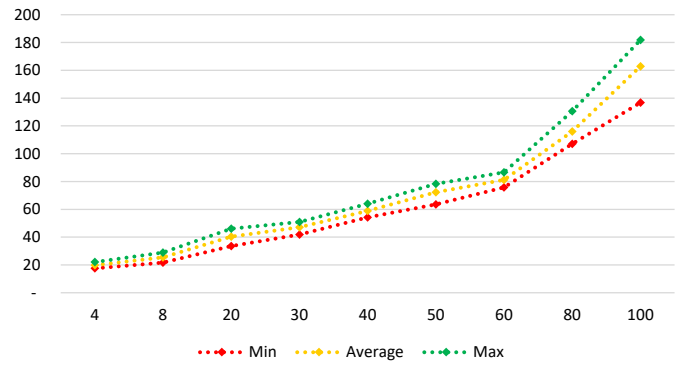
Exhibit 43 - Powerball relationship between entries, combinations and win probabilities (No. mlns, LHS; jackpot size A$m, x-axis) - pre Game Change



Source: Company Data, E&P Estimates.

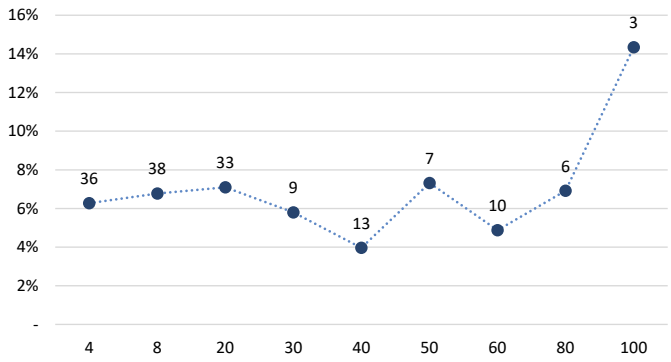
There is a potential for error from the historical data not being representative of the likely number of entries for a given draw size. Below we illustrate that within most of the draws that will occur, the margin of error on estimating total entries looks relatively small (i.e. basically the area between the yellow line and either the red/green lines) compared to the magnitude of the lines themselves. We note that the $100m draw has the highest standard deviation/average, but that this jackpot has too few observations to draw any concrete conclusions. We do suspect the lower jackpots to have greater consistency given the impact of subscription entry volumes (i.e. players automatically subscribed to purchase 'x' number of tickets at jackpots 'y' size), noting that $100m jackpots attract many one-time players.

Exhibit 44 - Spread of Powerball Entries (m's, y-axis) based on Jackpot (A$m, x-axis)



Source: Company Data, E&P Estimates. Note: Entries are from Jul-20

Exhibit 45 - Standard Deviation/Average (y-axis) based on Jackpot (A$m, x-axis) - jackpot count as label



Source: Company Data, E&P Estimates. Note: Entries are from Jul-20

Steps towards estimating FY24 sales from FY23

The steps toward estimating FY24 Powerball sales can be put into two buckets:

Known Factors

1. Weeks in the period. Both FY24 & FY23 have 52 Powerball draws, unlike FY22 which had 53.
2. Starting Jackpot. FY24 has a starting jackpot of $12m vs. only $4m in FY23 ($60m in FY22).
3. Game Change (i.e. Price Increase). FY24 will have a $1.2 subscription price for the entire period, vs. only 6 weeks in FY23

4. Prize Fund Allocation. FY24 will have marginally a lower division-one allocation (33.5% vs. 35%) for the same period as above (note we have split this from the game change, as the prize fund allocations can be altered without a full game change).

Estimates

1. Reversion to 50th percentile. We estimate FY23 sales were in the 51st percentile.
2. Sequence Impacts (i.e. Acceleration) & Beginning Prize Fund. We estimate that the sequence will remain slightly altered ($12m instead of $8m & $80 skipped straight to $100m). We also estimate the accelerated sequence will be used more frequently. This is despite our estimate that the beginning PRF is lower than it was in FY23 (needed in order to know when to first accelerate).
3. Like-for-Like Entries increases/decreases & Cannibalization. We estimate a -3% reduction in LFL entries. Part of this reduction is attributable to increased cannibalisation from Oz Lotto. We also factor in a premium for $100m+ draws and the potential impact of jackpot fatigue.

Assumptions for our model

1. Weekly variation in Entries**.** We assume constant entries throughout the entire simulation despite potential inflation impacts and the belief that there are likely factors that drive entry levels to deviate from the mean in a non-random manner such as seasonality, cannibalisation and macroeconomic effects on discretionary spending. The exception to this is jackpot fatigue, as we do penalize entries for $100m+ draws if there are multiple in a short time horizon.
2. Management Intervention. We do not believe there is a set formula for when management will accelerate and slow down the sequence. Our model assumes constant inputs to determine which sequence to use, despite our belief that management can and have in the past altered the sequence with levels of discretion instead of purely formulaically.
3. Player Behavior. We believe players will play certain numbers of Powerball in a non-random manner (e.g. birthdays). We do not account for this or any impact of ticket purchasing options (e.g. PowerHits, Syndicates, etc) when determining the expected number of unique entries for a given draw.
4. Finishing Jackpot. The primary result of our simulation is the overall sales. It is feasible that two simulations finish with the same sales, but have different finishing jackpots. In this instance, the higher finishing jackpot would be superior in the long run. However, our simulation only considers the recorded sales in the active period.

Monte Carlo Simulation Results

We estimate that a 50th percentile result in for Powerball sales in FY24 would be $3.3bn. This would represent a +5.4% growth vs. FY23 and +1.9% vs. the record high FY22. If LFL entries remain constant, we would expect growth of 10.9% at the 50th percentile level.

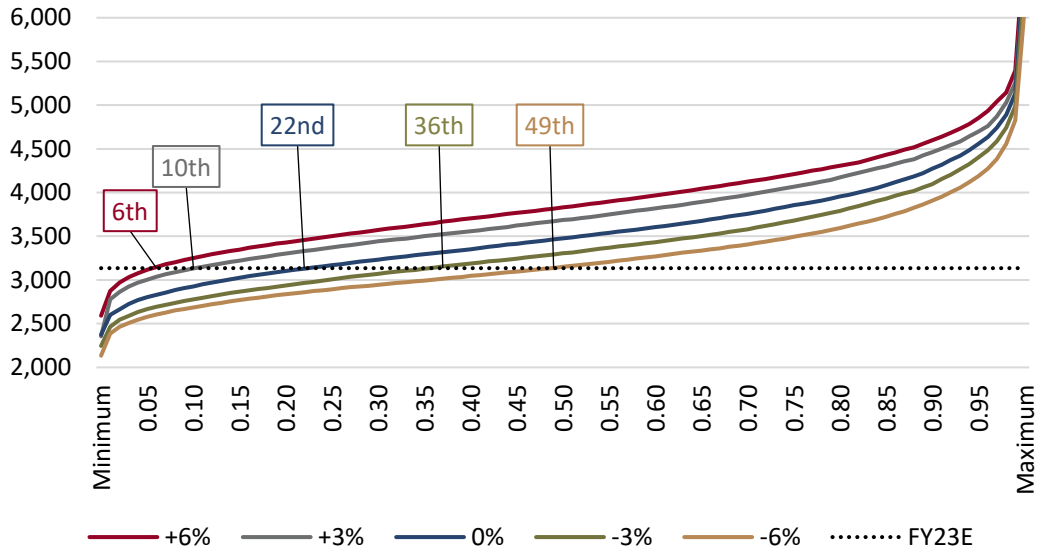
Exhibit 46 - FY24E Powerball full year monte carlo simulation sensitivity table based on LFL entries growth - sales ($bn.)

	Percentile										
	Min	1st	5th	10th	25th	50th	75th	90th	95th	99th	Max
+6%	2,592	3,251	3,429	3,571	3,703	3,830	3,968	4,126	4,305	4,597	6,887
+3%	2,356	3,132	3,301	3,442	3,557	3,685	3,821	3,973	4,174	4,461	6,589
-	2,374	2,926	3,105	3,234	3,352	3,476	3,604	3,758	3,955	4,274	6,693
-3%	2,245	2,778	2,937	3,070	3,190	3,305	3,431	3,581	3,791	4,096	6,425
-6%	2,134	2,687	2,836	2,944	3,046	3,151	3,271	3,407	3,595	3,908	6,119

Source: Company Data, E&P Estimates. Note: includes WA & retailer commission

We estimate that it would take a 22nd/36th percentile result with -3%/0% LFL entries growth for FY24 Powerball to achieve the same sales as FY23.

Exhibit 47 - FY24E Powerball full year monte carlo simulations based on LFL entries growth - sales ($bn.). Labels show percentile to equal FY23E sales.



Source: Company Data, E&P Estimates

Game change impacts

By running identical simulations and isolating individual inputs, we can estimate their magnitude:

1. Subscription Price. We estimate this has a +19.0% impact given how much it helps accumulate the PRF.
2. Commission Increase. Increase of +1.81% (the increase on ticket price). This doesn't impact the PRF and is therefore just subscription price x commission calculation. Only impact would be downward pressure on entries.
3. Prize allocation changes. We estimate this has a -3.5% impact.
4. Like-for-Like entries. We estimate this has a -4.4% impact.
5. Week 2 Jackpot change ($12m vs. $8m). We estimate this has a +2.8% impact.

July update

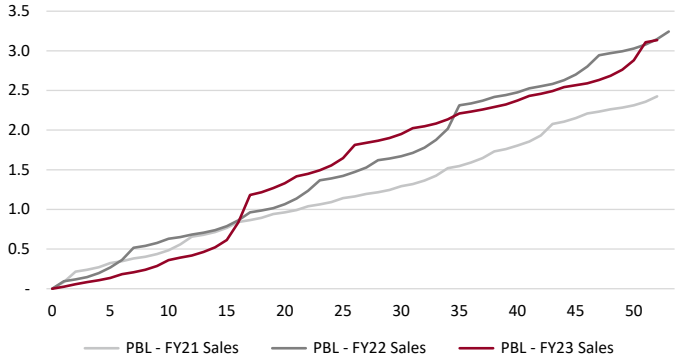
We note that as of the date of this report, zero Powerball jackpots have been won and the current jackpot is $100m. Consequently, our forecast has since been upgraded to the year-to-date actual sales, plus the 50th percentile sale for the remainder of the period.

FY23E

Powerball had a terrible start to the year in its first 10 weeks with the Div-1 prize reaching $40m only once. The accelerated sequence and favorable outcomes led to the $160m draw (the largest in Aus. history) on October 27 pulling 1H23 out of its slump and ending up with record half-yearly sales. To put the impact of the $160m draw into perspective, that draw alone recorded more sales than the first 9 weeks combined. Powerball then had another flat period from January to May, only going as high as $40m twice in the 21 weeks, before recovering with a $100m draw in the final sequence. Overall, we estimate Powerball sales fell -4.1% yoy.

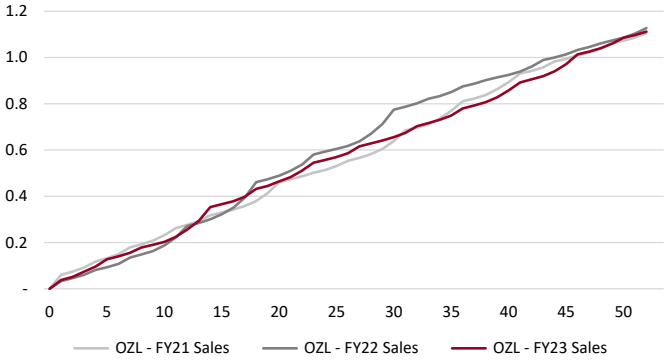
Oz Lotto struggled to gain any traction all year and was -1.6% down on FY22. This was unexpected given the theoretical benefits of the May-22 game change.

Exhibit 48 - Cumulative Powerball Sales (A$bn, y-axis) by week in year (x-axis)



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 49 - Cumulative Oz Lotto Sales (A$bn, y-axis) by week in year (x-axis)

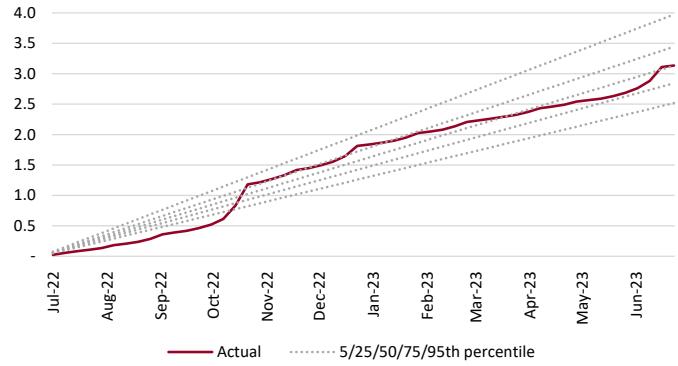


Source: Company Data, E&P Estimates. Note: Includes WA

We estimate FY23 PBL sales to be in the 51st percentile. 1H23 was strong and finished in the 77th percentile, aided by the $160m jackpot. 2H23 largely offset this and performed in the 35th percentile. Note that when back-testing, we account for the frequency of acceleration and the relevant period parameters (starting jackpot, number of weeks etc). Hence, a 50th percentile run in 1H23 does not necessarily equal a 50th percentile run in 2H23.

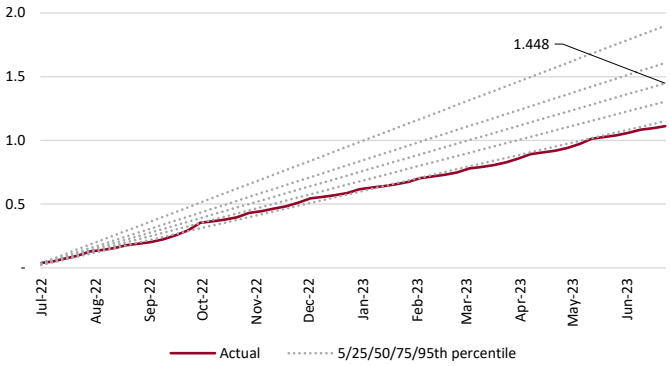
We estimate that FY23 OZL sales were in the 3rd percentile (ie. absolutely shockingly unlucky for TLC). Alternatively, sales would have been in the 23rd percentile if comparing the simulations run with the pre-game change settings. We still see the game change as a positive, particularly the lengthening of the division-one odds which should theoretically result in larger jackpots more frequently. We believe that should a reversion to the 50th percentile materialise, Oz Lotto could see 30.3% upside.

Exhibit 50 - Powerball Sales (A$bn, FY23)



Source: Company Data, E&P Estimates. Note: Includes WA.

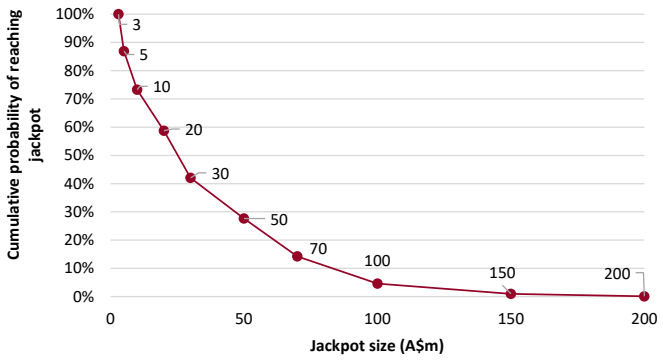
Exhibit 51 - Oz Lotto Sales (A$bn, FY23)



Source: Company Data, E&P Estimates. Note: Includes WA.

Whilst estimating Oz Lotto to have been in the 3rd percentile may seem extreme, the unfavorable sequences justify this. There were thirteen Oz Lotto sequences in FY23 and only one jackpot of $50m or greater, despite each sequence having a 27.6% chance to reach this size. Furthermore, six out of the ten $20m jackpots went off despite only a 28.4% chance to do so and four out of five $30m jackpots despite only a 34.3% chance (both for the individual draws).

Exhibit 52 - Cumulative Probability of reaching Oz Lotto Jackpot



Source: Company Data, E&P Estimates

Consequently, the actual weeks spent at each jackpot was far inferior to the theoretical average.

Exhibit 53 - FY23 Actual vs. Theoretical Oz Lotto Results

Weeks spent at each jackpot - FY23											
	3	5	10	20	30	50	70	100	120	150	Total
Actual	13.0	12.0	11.0	10.0	5.0	1.0	0.0	0.0	0.0	0.0	52.0
Theoretical avg.	12.7	10.8	8.9	7.0	5.9	3.9	1.9	0.6	0.1	0.0	52.0
Net	+0.3	+1.2	+2.1	+3.0	-0.9	-2.9	-1.9	-0.6	-0.1	-0.0	-

Sales at Each Jackpot - FY23											
	3	5	10	20	30	50	70	100	120	150	Total
Actual	162.9	183.6	217.5	299.2	187.5	59.5	0.0	0.0	0.0	0.0	1,110.2
Theoretical avg.	159.4	165.5	176.6	210.4	222.8	228.9	195.0	85.6	25.8	3.1	1,473.2
Net	+3.4	+18.1	+40.8	+88.9	-35.2	-169.5	-195.0	-85.6	-25.8	-3.1	-363.0

Source: Company Data, E&P Estimates. Note: theoretical avg. varies from 50th percentile.

FY23 Estimates by Game

The table below shows our forecasts for FY23E and a brief comparison vs. FY22:

Powerball. FY22 was an above average year which we estimate was in the 69th percentile. It also had an extra week and a starting jackpot of $60m vs. $4m.

Oz Lotto. Had a very unfavorable run as discussed above despite the game change.

Saturday Lotto. Very similar sales to FY22.

Monday & Wednesday Lotto. Marginal drop on FY22, but a fair way below FY21.

Set for Life. Small drop after record FY22.

Lucky Lotteries. Strong growth after a very good Superdraw run that had the jackpot reach $33.045m (theoretical average of $9.34m).

Exhibit 54 - E&P FY23E Turnover by Game (A$m)

Game	FY22A	FY23E	yoy%
Powerball	2,587	2,481	-4.1%
Oz Lotto	897	883	-1.6%
Saturday Lotto	1,752	1,777	1.4%
Monday & Wednesday Lotto	434	425	-2.0%
Set for Life	341	329	-3.4%
Lucky Lotteries	111	133	19.9%
Instants	441	468	6.1%
Other	153	164	7.2%
Total	**6,716**	**6,661**	**-0.8%**

Source: Company Data, E&P Estimates. Note: excludes WA

Financial Forecasts

We have launched a new TLC model as part of our coverage re-launch. Our model is built from the following broader operating assumptions:

- Lotteries FY23 revenues estimated using entries analysis for full year
- Lotteries FY24 revenues onwards modelled using a 50th percentile expected outcome approach from our monte carlo simulation efforts, and then longer term 5% growth rate;
- 58.7% long-term prize payout ratio (assumes a certain mix of games);
- 76.5% average taxes on return to game;
- Digital penetration grows 0.75% per 6mth period
- Service Fees and Other revenue grows 3.5% per annum;
- Opex growth ~4.0% per annum
- Keno revenue growth 4.0% per annum;
- Taxes, Commissions and other variable costs 45.0% of revenue (55% variable contribution margin);
- Opex growth 4.0% per annum

Our estimates are shown below, as well as changes vs last published.

Exhibit 55 - TLC changes to forecasts and valuation

		FY23	FY24	FY25
Revenue	New	3531.0	3843.5	3943.7
	Old	3357.0	3549.0	3692.0
	% change	5.2%	8.3%	6.8%
EBITDA	New	727.3	837.6	866.0
	Old	680.0	744.0	793.0
	% change	7.0%	12.6%	9.2%
EBIT	New	627.2	736.7	766.9
	Old	596.0	659.0	707.0
	% change	5.2%	11.8%	8.5%
NPAT	New	319.3	432.8	459.0
	Old	362.0	409.0	444.0
	% change	-11.8%	5.8%	3.4%
EPS (cps)	New	14.3	19.4	20.6
	Old	16.2	18.3	19.9
	% change	-11.5%	6.3%	3.6%

Source: E&Pe

Our forecasts vs Visible Alpha consensus are shown below.

Exhibit 56 - E&Pe vs Visible Alpha consensus

		FY23	FY24	FY25
Revenue	E&Pe	3531.0	3843.5	3943.7
	Consensus	3581.0	3848.8	4003.0
	% difference	-1.4%	-0.1%	-1.5%
EBITDA	E&Pe	727.3	837.6	866.0
	Consensus	738.7	807.7	850.3
	% difference	-1.5%	3.7%	1.8%
NPAT	E&Pe	319.3	432.8	459.0
	Consensus	288.7	402.7	432.8
	% difference	10.6%	7.5%	6.1%

Source: E&Pe, Visible Alpha

This table goes to the Neutral recommendation initially. For FY23 we are a marginally worried about downgrade risk (albeit -1.5% EBITDA misses are typically not a massive deal for a company like TLC). From FY24 onwards where our theoretical view of revenues start, we are more positive on long term margins, we believe due to our view on the recent retail margin changes, however all these figures are reliant on a theoretical 50th percentile scenario playing out. As such for now we are Neutral until we believe consensus has re-based correctly.

FY24 Estimates by Game

Our estimates for FY24 are:

Powerball. Strong uplift from the game change in conjunction with a 'perfect' start to the year (i.e. zero jackpots, current jackpot of $100m).

Oz Lotto. A reversion to 50th percentile.

Saturday Lotto. Small uplift from an extra week, offset by some cannibalization from Powerball & Oz Lotto gains.

Monday & Wednesday Lotto. Increase from an anticipated game change. Details and date are unknown at present.

Set for Life. Another marginal decline.

Lucky Lotteries. A decline after a decent year. The hardest game to estimate given its format.

Stock Focus

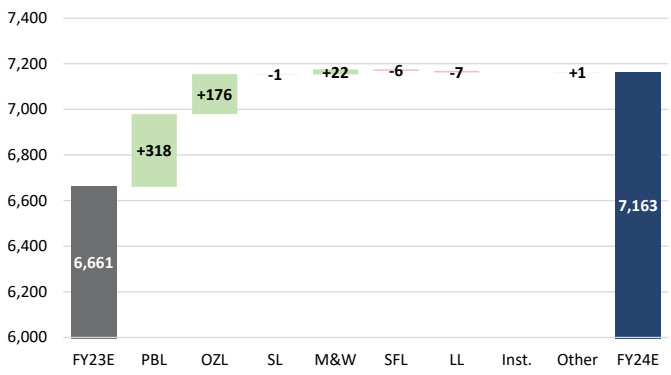
The Lottery Corporation

Table 1 - E&P FY24E Turnover by Game (A$m)

Game	FY23E	FY24E	yoy%
Powerball	2,481	2,800	12.8%
Oz Lotto	883	1,058	19.9%
Saturday Lotto	1,777	1,776	-0.1%
Monday & Wednesday Lotto	425	447	5.2%
Set for Life	329	323	-1.8%
Lucky Lotteries	133	126	-5.5%
Instants	468	468	-
Other	164	165	0.5%
Total	**6,661**	**7,163**	**7.5%**

Source: Company Data, E&P Estimates. Note: excludes WA

Chart 1 - FY23-FY24 Lotteries sales waterfall - game impacts (A$m)



Source: Company Data, E&P Estimates. Note: excludes WA

Valuation

Our valuation of TLC is $5.79/sh. It is derived from the following key assumptions:

- Financial forecasts to FY33;
- Cost of Equity 11.0% (5% risk free rate, 6% market risk premium, 1.0 beta);
- Terminal EV/EBITDA multiple 17.0x in line with present trading multiple;
- 6.00% long term cost of debt
- Victorian, Tasmanian and Northern Territory concessions are renewed at same terms as last deal during the forecast period.

Concession terms are outlined in the table below generally, as well as last one-time payments made. For NSW, QLD and SA we believe the concessions are long-dated enough that this is unlikely to materially impact the TLC trading multiple in the forecasting period even though the theoretical asset value of the concession asset will have reduced (the magnitude of the concession payments vs the value indicates to us an indirect nexus rather than direct linear value recovery by state governments).

Table 2 - TLC lottery licenses

State		NSW	VIC	QLD	SA	TAS	NT
License Date	Year	2010	2018	2007	2012	2020	2012
End Date	Year	2050	2028	2072	2052	2025	2032
Period	Yrs	40	10	65	40	5	20
Exclusivity	Yrs	40	10	9	40	5	20
Fee	$m	850	120	530	430	0	10
Years remaining	Yrs	27	5	49	29	2	9

Source: Company data, E&P estimates. Note: Tasmania license fee unknown.

Stock Focus

The Lottery Corporation

The Lottery Corporation, Neutral, VALUATION AUD5.79



EAP CORE DRIVERS

- Household disposable income and population growth driving gambling spend.
- Product innovation in lotteries games.
- Online migration impacting revenue growth and margins.

INVESTMENT THESIS

- Sales growth driven by: (i) strong customer acquisition during Covid-19; and (ii) recent and future game changes.
- Higher margins as sales migrate online, with online sales EBITDA margins more than double that of retail.
- Capital management initiatives given significant FCF generation whilst being under the target gearing range.

Stock Focus

The Lottery Corporation

FINANCIAL SUMMARY

The Lottery Corporation			TLC-ASX		
As at:	03/08/2023		Recommendation:	Neutral	Share Price: AUD5.23

Year end	Jun	2022A	2023E	2024E	2025E
INCOME STATEMENT					
Total Operating Income/Revenue		3,278.6	3,531.0	3,843.5	3,943.7
EBITDA		606.1	727.3	837.6	866.0
D&A		(61.9)	(100.2)	(100.8)	(99.1)
EBIT		544.2	627.2	736.7	766.9
Net Interest Expense		(12.1)	(117.6)	(118.4)	(111.2)
PBT		522.4	460.4	618.3	655.8
Tax expense		(179.7)	(141.1)	(185.5)	(196.7)
NPAT attributable to s'hers		**342.7**	**319.3**	**432.8**	**459.0**
Adjusted Net Profit		342.7	319.3	432.8	459.0
Year end	**Jun**	**2022A**	**2023E**	**2024E**	**2025E**
BALANCE SHEET					
Assets					
Current Receivables		47.9	47.8	56.4	58.7
Other Current Assets		342.6	422.0	422.0	422.0
Total Current Assets		706.7	724.3	960.9	1,129.0
Property Plant & Equipment		156	136	131	128
Intangibles		2,981.3	2,956.9	2,928.3	2,901.3
Other Non-Current Assets		284.0	246.6	246.6	246.6
Total Non-Current Assets		3,510.6	3,392.5	3,359.0	3,328.9
Total Assets (BS)		4,217.3	4,116.8	4,319.9	4,457.9
Liabilities					
Total Current liabilities		3,355.5	3,185.8	3,312.1	3,347.2
Total Non-Current Liabilities		619.0	591.2	591.2	591.2
Total Liabilities		3,975	3,777	3,903	3,938
Total Shareholders' Equity		242.8	339.8	416.5	519.4
Net Assets		242.8	339.8	416.5	519.4
Net Tangible Assets		(2,738.5)	(2,617.1)	(2,511.8)	(2,381.9)
Book Value		242.8	339.8	416.5	519.4
Year end	**Jun**	**2022A**	**2023E**	**2024E**	**2025E**
OPERATING CASH FLOWS					
EBITDA		787.0	845.1	837.6	866.0
Change in Working Capital		0.0	(174.9)	117.7	32.8
Gross Operating Cash Flow		**787.0**	**670.2**	**955.3**	**898.9**
Net interest paid		(5.7)	(116.3)	(118.4)	(111.2)
Tax Paid		(156.9)	(130.0)	(185.5)	(196.7)
Net Operating Cash Flows		**624.4**	**423.8**	**651.4**	**590.9**
Investing Cash Flows					
Purchase of PP&E		(24.4)	(49.6)	(67.3)	(69.0)
Other Investing Cashflows		(39.1)	(26.9)	0.0	0.0
Total Investing CashFlow		**(63.5)**	**(76.5)**	**(67.3)**	**(69.0)**

Source: Company data, E&P estimates

Year end	Jun	2022A	2023E	2024E	2025E
FINANCING CASH FLOWS					
Proceeds from Borrowings		(455.0)	(220.0)	0.0	0.0
Net Proceeds Issue/Buyback of Securities		0.0	(6.3)	0.0	0.0
Distributions		0.0	(178.1)	(356.1)	(356.1)
Total Financing Cash Flow		**(456.6)**	**(409.1)**	**(356.1)**	**(356.1)**
Net Change in Cash		104.3	(61.7)	228.0	165.8
Free Cashflow		600.0	374.2	584.1	521.9
Year end	**Jun**	**2022A**	**2023E**	**2024E**	**2025E**
PER SHARE DATA					
EPS - Adjusted (cents)		15.4	14.3	19.4	20.6
DPS (cents)		0.0	0.2	0.2	0.2
Cash Flow Per Share (cents/share)		27.0	16.8	26.2	23.4
Book Value ($/sh)		10.9	15.3	18.7	23.3
NTA Per Share ($/sh)		(123.0)	(117.6)	(112.8)	(107.0)
Year end	**Jun**	**2022A**	**2023E**	**2024E**	**2025E**
SEGMENTALS					
Revenues		3,272.8	3,116.5	3,300.9	3,436.1
Revenues		244.5	240.1	247.8	255.8
Revenues		625.3	591.9	653.4	699.7
Revenues		85.4	87.6	90.5	93.5
Year end	**Jun**	**2022A**	**2023E**	**2024E**	**2025E**
ANALYSIS					
EBITDA Margin (%)		18.5	20.6	21.8	22.0
EBIT Margin (%)		16.6	17.8	19.2	19.4
Net Profit Margin (%)		10.5	9.0	11.3	11.6
Capital Employed		2,315.8	2,285.0	2,133.7	2,070.8
Return on Capital Employed (%)		45.8	27.3	33.3	36.5
EBITDA/Capital Employed (%)		51.0	31.6	37.9	41.2
ROE %		**133.1**	**109.6**	**114.5**	**98.1**
Net Debt/Equity (%)		853.8	572.4	412.2	298.7
Net Debt/Net Debt & Equity (%)		89.5	85.1	80.5	74.9
Net Debt/EBITDA (x)		3.4	2.7	2.1	1.8
Net Interest Cover (x)		45.0	5.3	6.2	6.9
GOCF/EBITDA (%)		1.0	0.8	1.1	1.0
Total Capex/Sales (%)		0.7	1.4	1.8	1.8
Working Capital/Sales (%)		(0.2)	(0.2)	(0.2)	(0.2)
D&A/PP&E (%)		39.7	73.4	76.7	77.2
EV/EBITDA (x)		22.6	18.7	15.9	15.2

BLENDED VALUATION		
Valuation - DCF	5.79	100.0%
Valuation - Capitalisation of Earnings	0.00	0.0%
Valuation - PER	0.00	0.0%
Valuation - SOTP	0.00	
Valuation	AUD5.79	

Stock Focus

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RESEARCH RECOMMENDATION DEFINITIONS

Positive	Stock is expected to outperform the S&P/ASX 200 over the coming 24 months.
Neutral	Stock expected to perform in line with the S&P/ASX 200 over the coming 24 months.
Negative	Stock is expected to underperform the S&P/ASX 200 over the coming 24 months.
Speculative Buy	Stock has limited history from which to derive a fundamental investment view or its prospects are highly dependent on event risk, e.g. Successful exploration, scientific breakthrough, high commodity prices, regulatory change, etc. Consequently, the stock is considered a high-risk investment which may be prone to high volatility in share price movements, have a greater risk of capital loss and/or the stock may have low liquidity.
Suspended	Stock is temporarily suspended due to compliance with applicable regulatory and/or E&P policies in circumstances where E&P is acting in an advisory capacity.
Not Rated	Stock is not included in our investment research universe.

Research Criteria Definitions

Recommendations are primarily determined with reference to how a stock ranks relative to the S&P/ASX 200 on the following criteria:

Valuation	Composite of Rolling 12-month prospective multiples and discounted cash flow (DCF), or DCF for resource stocks.
Earnings Outlook	Forecast 2-year EPS growth.
Earnings Momentum	Percentage change in the current consensus EPS estimate for the stock (rolling 1-year forward basis) over the consensus EPS estimate for the stock 3 months ago.
Shareholder Returns	Composite of forecast ROE (rolling 1-year forward basis) and the percentage change in ROE over 2 years.
Debt Servicing Capacity	Rolling 12-month EBIT Interest Cover ratio.
Cyclical Risk	Qualitative assessment of the 2-year outlook for a stock/industry's profit cycle.
Industry Quality	Qualitative assessment of an industry's growth/returns potential and company specific management capability.
Financial Transparency	If we don't understand it, we won't recommend it.

For stocks where Evans and Partners does not generate its own forecasts, Bloomberg consensus data is used. Analysts can introduce other factors when determining their recommendation, with any material factors stated in the written research where appropriate.

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Stock Focus

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